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Unveiling the influence of microaeration and sludge recirculation on enhancement of pharmaceutical removal and microbial community change of the novel anaerobic baffled biofilm - membrane bioreactor in treating building wastewater

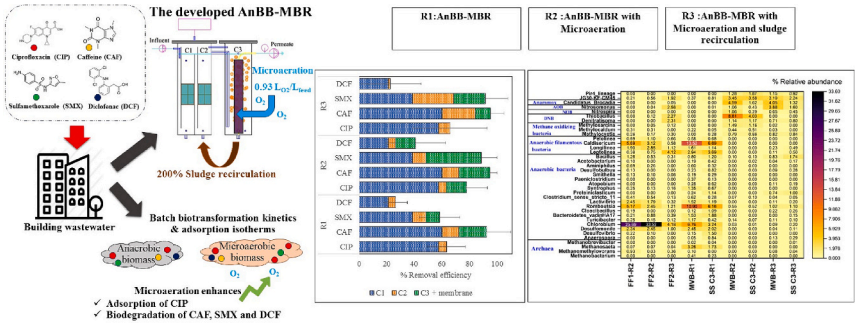
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HIGHLIGHTS

- A novel AnBB-MBR with microaeration and sludge recirculation was proposed.
- Influence of integrated microaeration and sludge recirculation was investigated.
- Microaeration could enhance CIP adsorption and biodegradation of CAF, SMX and DCF.
- Sludge recirculation improved SMX removal performance of the microaerated AnBB-MBR.
- Methane-oxidizing bacteria that can degrade SMX were abundant in microaerobic zone.

GRAPHICAL ABSTRACT



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ABSTRACT

This research aims to conduct a comparative investigation of the role played by microaeration and sludge recirculation in the novel anaerobic baffled biofilm-membrane bioreactor (AnBB-MBR) for enhancing pharmaceutical removal from building wastewater. Three AnBB-MBRs — R1: AnBB-MBR, R2: AnBB-MBR with microaeration and R3: AnBB-MBR with microaeration and sludge recirculation — were operated simultaneously to remove Ciprofloxacin (CIP), Caffeine (CAF), Sulfamethoxazole (SMX) and Diclofenac (DCF) from real building wastewater at the hydraulic retention time (HRT) of 30 h for 115 days. From the removal profiles of the targeted pharmaceuticals in the AnBB-MBRs, it was found that the fixed-film compartment (C1) could significantly reduce the targeted pharmaceuticals. The remaining pharmaceuticals were further removed with the microaeration compartment. R2 exhibited the utmost removal efficiency for CIP (78.0 %) and DCF (40.8 %), while SMX was removed most successfully by R3 (microaeration with sludge recirculation) at 91.3 %, followed by microaeration in R2 (88.5 %). For CAF, it was easily removed by all AnBB-MBR systems (>90 %). The removal mechanisms indicate that the microaeration in R2 facilitated the adsorption of CIP onto microaerobic biomass, while the enhanced biodegradation of CAF, SMX and DCF was confirmed by batch biotransformation kinetics and the adsorption isotherms of the targeted pharmaceuticals. The microbial groups involved in biodegradation of the

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targeted compounds under microaeration were identified as nitrogen removal microbials (Nitrosomonas, Nitrospira, Thiobacillus, and Denitratisoma) and methanotrophs (Methylosarcina, Methylocaldum, and Methylocystis). Overall, explication of the integration of AnBB-MBR with microaeration (R2) confirmed it as a prospective technology for pharmaceutical removal from building wastewater due to its energy-efficient approach characterized by minimal aeration supply.

1. Introduction

Recently, pharmaceuticals have been detected in the effluent from wastewater treatment plants (WWTPs) in the range of ng/L to μg/L as a result of WWTPs not being designed to remove these pharmaceuticals efficiently (Tran et al., 2018). The contamination of the water environment by pharmaceuticals could pose an ecological risk to aquatic and terrestrial environments in the short and/or long-term, such as from the occurrence of antibiotic resistant genes, reproductive problems, cytological alterations, and lethality. Moreover, human health can also be negatively impacted by the accumulation of pharmaceuticals in the food chain, particularly through the water consumed for agricultural and livestock farming (Fent et al., 2006).

Ciprofloxacin (CIP), Caffeine (CAF), Sulfamethoxazole (SMX) and Diclofenac (DCF) are the main pharmaceuticals commonly found in surface water and effluents of WWTPs in Thailand with a high hazard quotient remaining in the water basin reported by various studies (Sinthuchai et al., 2021; Tewari et al., 2013). CAF is a psychostimulant and analeptic drug used to alleviate drowsiness and lethargy (Frederic and Yves, 2014). CIP is a fluoroquinolone antibiotic that has an impact on aquatic organisms at concentrations in the range of a few hundred μg/L (Nguyen et al., 2019), and SMX is an oral sulfonamide antibiotic used for the treatment of various bacterial infections (Xu et al., 2022). DCF is a non-steroidal anti-inflammatory drug (NSAID), whose acute and chronic toxicity to aquatic organisms has been reported at a level of one to a hundred μg/L (Triebskorn et al., 2004).

Recently, anaerobic treatment has been widely incorporated into membrane bioreactor (MBR) to create the anaerobic membrane bioreactor (AnMBR), which has many benefits including in terms of total microbial sludge retention, high effluent quality, and high toxic wastewater tolerance. Most previous studies in the fields of pharmaceutical removal by anaerobic MBR have focused on completely stirred tank reactor (CSTR), upflow anaerobic sludge blanket (UASB), fluidized, submerged anaerobic MBR, etc. (Ji et al., 2020; Oberoi et al., 2022). Buakaew and Ratanatamskul (2021) developed inclined-tubes to be used in the fixed film system that offers high retention of the microbial sludges and promotes biofilm growth inside the combined AnBB-MBR, which could efficiently remove organics and nutrients from building wastewater, and the system could operate at a stable flux for long experimental runs. Furthermore, this system exhibits high biomass retention capabilities, which could accommodate many types of microorganisms, based on the abundance of baffled compartments that was supposed to be an efficient wastewater treatment system (Ratanatamskul and Katasomboon, 2017). In addition, the biofilm layer was reported as being able to prevent the toxic effects caused to microbes by toxic compounds, and to enhance the biodegradation of many pollutants (Carneiro et al., 2019).

It has been reported that microaeration with low amounts of oxygen up to 1 L_{O2}/L_{feed} and dissolved oxygen (DO) in the range of 0.1–1 mg/L is able to enhance the removal of volatile fatty acids (VFAs) and nitrogen in the anaerobic treatment process (Buakaew and Ratanatamskul, 2023). In addition, a low amount of oxygen has also been found to play a role in influencing the redox equilibrium of $NADH/NAD^+$, thereby stimulating the production of enzymes by acidogenic biomass and enhancing the rate of hydrolysis (Huiliñir et al., 2023). Another study reported that pharmaceutical removal in the anaerobic UASB system with a short HRT of 7.4 h, using microaeration equivalent to 0.025–0.152 $L.O_2/L_{feed}$ could also improve SMX and DCF removal (do Nascimento et al., 2021).

Furthermore, sludge recirculation has been commonly utilized in anoxic/oxic bioreactors (A/O) to mitigate the accumulation of nitrate (NO_3^-) for enhancing the effectiveness of nitrogen removal from building wastewater (Fathali et al., 2019). The sludge recirculation might also enhance the biodegradation of slowly biodegradable organic matters in the wastewater due to variations in the environment conditions (anoxic/oxic conditions). At present, there is little information on the effect of sludge recirculation on pharmaceutical removal in the microaerobic/anoxic biotreatment systems. Therefore, there is a need for research to investigate the effect of sludge recirculation in the microaerobic/anoxic compartments of AnBB-MBR in order to determine the possible impact on the removal of pharmaceuticals as well as on the membrane fouling of the system in treating domestic building wastewater contaminated with pharmaceuticals.

This work aims to develop novel microaerobic AnBB-MBR systems with microaeration at 0.1 L/min (0.93 L_{O2}/L_{Feed}) with and without sludge recirculation for enhancing the removal of the targeted pharmaceuticals (CIP, CAF, SMX and DCF) in the AnBB-MBR. At present, no research has focused on the removal of pharmaceuticals by this proposed system. In addition, the effectiveness of microaeration in the anaerobic process of AnBB-MBR should be more clearly understood in terms of how microaeration affects the pharmaceutical removal mechanisms, the fate, biotransformation rate, and adsorption, as compared to anaerobic AnBB-MBR. Here, the AnBB-MBR systems were developed for three distinct bioreactors — an AnBB-MBR system (without microaeraton), an AnBB-MBR with microaeration, and an AnBB-MBR system incorporating microaeration and sludge recirculation — in order to gain insights into how a low airflow (0.1 L/min) and sludge recirculation could affect the pharmaceutical fate and removal. At present, no comparative study has been conducted on the effects of microaeration and microaeration plus sludge recirculation in AnBB-MBR on enhancing pharmaceuticals. Also, the comparative study of biotransformation kinetic rates and adsorption of the targeted compounds between microaerobic and anaerobic AnBB-MBR intensively investigated the different behaviors of anaerobic and microaerobic biomasses (biofilms and suspended sludges) on pharmaceutical removal in two batch experiments for active and inactive biomasses. This work also examined the organic removal performances of the systems, specifically focusing on parameters such as VFAs, and dissolved methane organic constituents. Additionally, the biomass characteristics, membrane fouling tendencies, and microbial community structure of the three AnBB-MBR systems were investigated in order to gain a comprehensive understanding of the system performance.

2. Materials and methods

2.1. Three AnBB-MBR configurations

As shown in Fig. 1, three AnBB-MBRs were constructed with identical dimensions of 60 cm in height, 45 cm in length, and 15 cm in width. Each system was divided into three compartments using baffles. The volume ratios of the first compartment (C1), second compartment (C2), and third compartment (C3) were 11.25 L, 11.25 L, and 18 L, respectively. Both C1 and C2 were filled with two sets of fixed films made of fiber glass filter (Glass filter mat, AQUA-NEAT, Japan). These filter sets had a specific area of 500 $m^2.m^{-3}$ and a total volume of 4.5 L to capture and retain the biomass by promoting the growth of biofilm.

Furthermore, C3 of the three bioreactors were filled with moving bed (MVB) media at a volume ratio of 30 % (Vmedia:Vcompartment). The MVB media was made of polyethylene with a circular length of 2.5 cm, a thickness of 0.11 cm and an active surface area of 4850 $m^2.m^{-3}$. The proposes of this MVB were to promote biofilm growth, facilitate membrane scouring and provide mixing in the C3 zone by using a sonic pump (AP1600). Each C3 of the three bioreactors contained a ceramic flat sheet microfiltration membrane (Ceraflo, Singapore) with a pore size of 0.5 μm and an average surface area of 0.107 m^2. A vacuum pressure gauge was connected to the membrane to monitor the transmembrane pressure (TMP). Feeding, permeates and sludge recirculation were operated by metering pumps (Prominent, Germany). Level sensors were applied to control the wastewater level inside the bioreactors. In this study, the first bioreactor was a control reactor of anaerobic baffled biofilm-MBR (R1: AnBB-MBR). The second bioreactor was the AnBB-MBR with microaeration (R2: AnBB-MBR with microaeration) using an air flowrate of 0.1 L_{air}/min in C3 by controlling the DO using a DO controller (Ponpe Instruments, Thailand) corresponding to DO in the range of 0.3–0.7 mg/L and oxygen supply rate at 0.93 L_{O2}/L_{feed}. The third bioreactor was the AnBB-MBR with the same microaeration as the second bioreactor, and sludge recirculation at 200 % was conducted from C3 to C2 (R3: AnBB-MBR with microaeration and sludge recirculation) to promote the anoxic/microaerobic conditions. This anoxic condition might enhance the biodegradation of SMX (Li et al., 2023b). The measured DO concentrations in C3 throughout the experimental runs were in the range of 0–0.2 mg/L, 0.3–0.7 mg/L and 0.3–0.7 mg/L for R1, R2 and R3, respectively (Supplementary data Fig. S1).

2.2. Wastewater, pharmaceuticals, and experimental run

High-rise building wastewater from the toilets and canteen inside Mahitaladhibesra Building, Chulalongkorn University was used as the domestic wastewater in this study. The characteristics of the wastewater consisted of chemical COD 245.80 ± 94.18 mg/L, TKN 88.43 ± 6.4 mg/

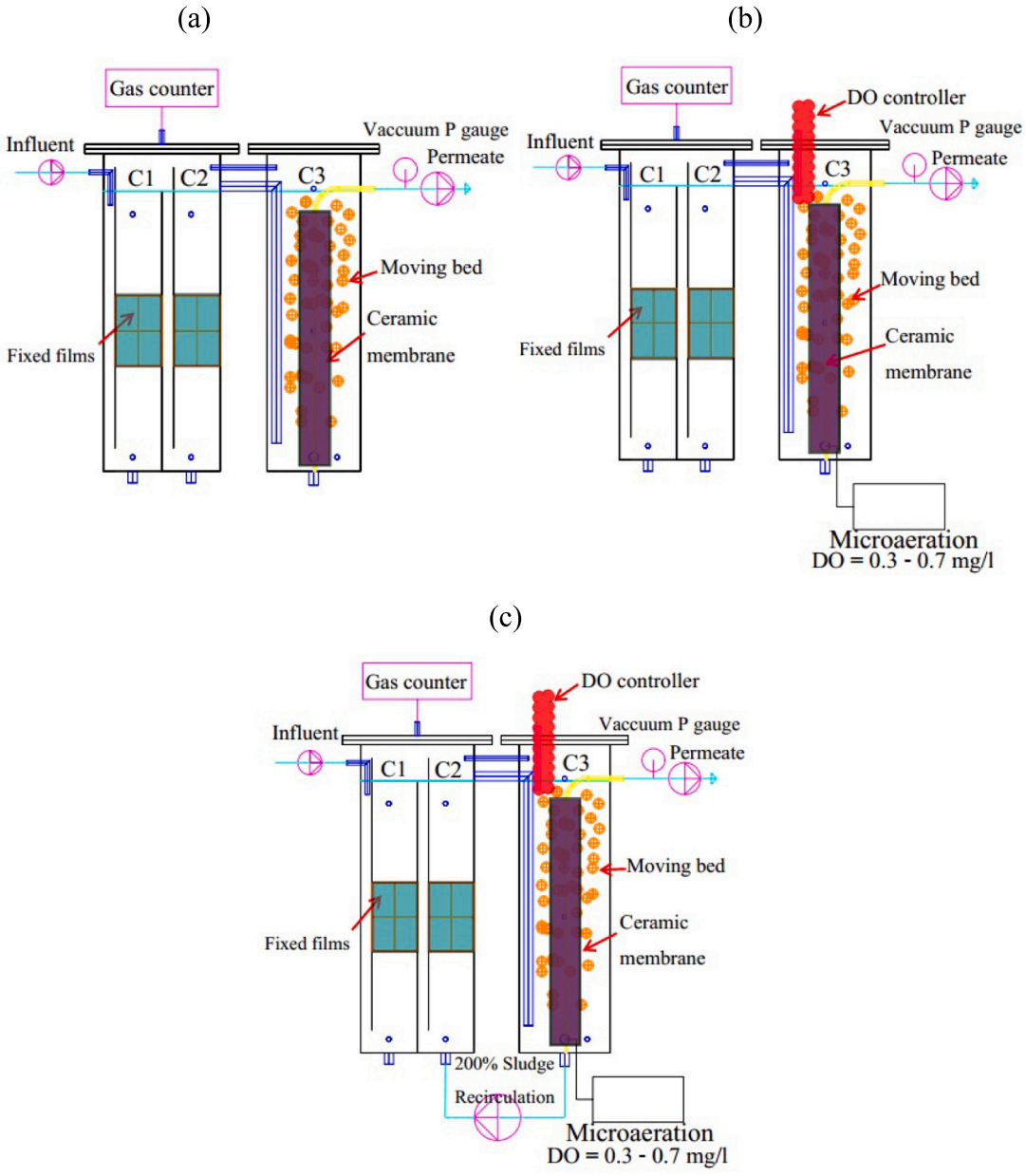


Fig. 1. Schematic diagram of three AnBB-MBRs: (a) R1: AnBB-MBR, (b) R2: AnBB-MBR with microaeration and (c) R3: AnBB-MBR with microaeration and sludge recirculation.

L, NH_4^+-N 83.30 ± 20.10 mg/L, PO_4^{3-}-P 11.79 ± 5.98 mg/L, and pH 7.43 ± 0.08. All of the targeted pharmaceuticals (CIP, CAF, SMX and DCF) purchased from Sigma-Aldrich (USA) were separately prepared in an individual stock solution at 2 g/L and kept at −20 °C before spiking. The CIP was prepared in 0.1 % formic acid solution; The CAF was prepared in Milli-Q water and the SMX and DCF were prepared in pure methanol. For the experimental runs of the three bioreactors, all targeted compounds were continuously spiked in the influent tank (750 l) until they reached 250 μg/L since this value is higher than the concentration of pharmaceuticals detected in the influent of WWTPs globally (Tran et al., 2018). The anaerobic sludges from a textile WWTP that were filled into C1 and C2 of the three bioreactors at 30 % of reactor volume were continuously acclimated with high-rise building wastewater with the targeted pharmaceuticals over three months before starting the experimental run.

The three bioreactors were then simultaneously operated in experimental runs for 115 d at a solid retention time (SRT) 100 d (calculated from C3 compartment) and constant HRT of 30 h (optimal value from the previous pilot-scale testing), which was divided into an HRT of 8.33 h in each C1 and C2 and 13.33 h in C3 of the AnBB-MBRs. The permeate flow rate was controlled in an intermittent manner. Specifically, it involved 8 min of suction followed by 2 min of relaxation to avoid excessive membrane stress. The permeate fluxes of the three bioreactors were kept constant at 12.6 $L.m^{-2}.h^{-1}$ to evaluate the fouling rate by TMP evaluation. After the TMP reached 375 mmHg, the membranes were removed from the reactors for cleaning. The cleaning process involved scrubbing the fouling substances on the membrane surfaces and then immersing them in a solution of 0.4 % NaOCl for 8 h. Subsequently, the membranes were washed with tap water and soaked in 0.4 % nitric acid for another 8 h. Finally, they were rinsed again with tap water before being reused in the AnBB-MBRs.

2.3. Analytical methods

COD, mixed liquor suspended solids (MLSS) and mixed liquor volatile suspended solids (MLVSS) were analyzed using standard methods (APHA, 2017). To determine the biomass concentration of the biofilms on the MVBs, ten MVBs from C3 of each bioreactor were dried overnight at 60 °C, followed by weighing. To detach the biofilms from the MVBs, a series of washing steps were performed. Firstly, the dried biofilms were washed and scrubbed with 6 % NaOCl solution, followed by 0.4 % HNO_3 and then deionized water. Subsequently, the MVBs were once again dried overnight at 60 °C, and their weights were measured to determine the weight of the fresh MVBs in order to obtain the attached biofilms weights. The pH and ORP of the wastewater were analyzed using a pH/ORP meter - HI 98121 (Hanna instruments, Italy). The zeta potential of the biomass in C3 was measured using a Zetasizer Nano ZSP (Malvern, England). A particle size distribution of the sludges in C3 of each bioreactor was performed using a laser particle size distribution analyzer - Mastersizer 3000 (Malvern, England).

The surface morphology of the fresh and used MVBs were visualized using a Scanning Electron Microscope (SEM) JSM-IT500HR, JED-2300 (JEOL, USA). To extract soluble microbial products (SMP) and loosely and tightly bound extracellular polymeric substances (L-EPS and T-EPS) from the suspended sludges and the foulants on the membrane, the suspended sludge samples were analyzed for MLSS and then 25 mL were added into the centrifuge tubes while the foulants were weighed and subsequently 25 mL of deionized (DI) water was added and mixed vigorously. After mixing and vortexing for 1 min, the samples were centrifuged at 4000*g* for 10 mins resulting in the separation of the supernatants containing SMP. Next, the dewatered sludges were mixed with 25 mL of a 0.9 % NaCl solution, vortexed, and centrifuged again. The supernatants obtained from this step represented L-EPS. Lastly, the remaining dewatered sludges were resuspended in another 25 mL of a 0.9 % NaCl solution and incubated in a water bath at 60 °C for 1 h followed by vortexing and centrifuging again; the supernatants in this case were representative of T-EPS. The modified phenol-sulfuric acid method was employed to determine the carbohydrate content using glucose as a standard (BeMiller, 2017), and the modified Lowry method, with bovine albumin serum as a standard was used to assess the protein content (Markwell et al., 1978). The three dimensions of the fluorescence excitation-emission matrix (3D-FEEM) spectra of the influent and mix liquors in C3 in each bioreactor were examined by measuring the excitation (Ex) spectra between 200 and 550 nm and the emission (Em) spectra between 250 and 550 nm. The samples were filtered through a 0.45 μm filter, and a fluorescence spectrofluorometer (Aqualog®, HORIBA Scientific., USA) was used to analyze them with an integration time of the excitation wavelength of 3 min. SCD1 served as the detector, with a grating density of 1200 for excitation and 285 for emission wavelengths. Fourier-transform infrared spectroscopy (FTIR) — conducted using a Perkin Elmer Spectrum: One instrument with a wavelength range of 4000 to 400 cm^{-1} — was employed to examine the functional groups present in the organic membrane foulants. Prior to the FTIR analysis, the foulants were subjected to drying at 60 °C for 8 h to obtain dried samples for analysis.

Analysis of VFAs, including acetic acid, propionic acid, butyric acid, isobutyric acid, butyric acid, and isovaleric acid, was carried out using gas chromatography with a flame ionization detector (GC-FID) with a DB-WAX column (30 m × 0.25 mm, Agilent, USA). Calibration curves for all measurements were prepared using VFA standards (35,272, Restek) with concentrations ranging from 2 to 200 mg/L. The column was set at 90 °C, and the temperature was increased at a rate of 4 °C/min until it reached 140 °C. Subsequently, the temperature was increased at a rate of 25 °C/min up to 250 °C, and this temperature was maintained for 2 mins. The injector and detector temperatures were both set at 250 °C. Helium gas (99.99 %, Praxair, Canada) was used as the carrier gas at a pressure of 97.6 kPa. Prior to analysis, all samples underwent acidification with ortho-phosphoric acid to achieve a pH of 2–3. After acidification, the samples were centrifuged at 4500 rpm for 10 min, and the resulting supernatants were filtered using a nylon syringe filter (0.22 μm) before being injected into the GC system. Dissolved methane analysis in the permeate of each bioreactor was carried out using the headspace method modified by DelSontro et al. (2016). A 500 mL syringe was connected to a stopcock valve to collect 250 mL of the permeated tube followed by the injection of 250 mL of nitrogen gas into the headspace, after which, the stopcock valve was closed immediately. The syringe was then shaken for 5 min to allow for equilibrium between the liquid and gas phases based on Henry's law. The gas in the headspace was analyzed using a GC equipped with a thermal conductivity detector (GC-TCD). The dissolved methane concentration was determined using Eq. (1):

$$C_{D-CH4} = (C_{CH4} \times P \times K_{CH4} \times MW_{CH4}) + (C_{CH4} \times V_{Head} \times MW_{CH4} \times T_0) \times \frac{1}{\left(V_{Head} \times \frac{22.4L}{1mol} \times T_1\right)} \quad (1)$$

where $C_{D\text{-}CH4}$ is the concentration of dissolved methane in the permeate (mg/L), C_{CH4} is the percentages of methane in the headspace of the syringe, P is pressure inside the bioreactor, V_{Head} is the volume of headspace in the syringe (250 mL), MW_{CH4} is molecular weight of methane (16 g/mol), T_0 and T_1 are 273.15 K and 298.15 K, respectively and K_{CH4} is Henry's law constant at ambient temperature (mol/L-atm).

2.4. Analysis of targeted pharmaceuticals

For the extraction of targeted pharmaceuticals in the liquid phase, 5 mL of the samples were first filtered by nylon syringe filter 0.45 μm to eliminate suspended solids and then diluted to 100 mL with Milli-Q water. Solid-phase extraction (SPE) was performed using 6 mL, 200 mg Oasis HLB cartridges (Waters, Milford, MA, USA) equipped with a

vacuum manifold pump. The cartridges were initially conditioned with 5 mL methanol followed by 5 mL Milli-Q water, and the samples were immediately loaded at the flowrate of 2–3 drops/s. The cartridges were then washed with 5 mL of Milli-Q water and vacuum dried for 30 min. Next, the analytes were gently eluted with 5 mL of methanol. Subsequently, the eluents were dried with a gentle stream of nitrogen until they were close to dryness, after which they were reconstituted in 5 mL of 50 % methanol in Milli-Q water. Finally, the samples were filtered through a PP membrane syringe filter (0.22 μm) before analysis. The biomass samples (suspended sludges/biofilm) were freeze dried and ground to powder to enable extraction of the targeted compounds. The resulting powder weighed approximately 0.2–0.5 g before being transferred to a centrifuge tube, into which, 10 mL of 50 % methanol was added. The solvents were vortexed for a while and sonicated at 40 °C for 15 min. Subsequently, the solvents were centrifuged at 4000 rpm for 10 min, and the supernatants were collected in the beakers. The extracted sludges were repeatedly mixed with 10 mL of 50 % methanol and extracted by the same mentioned procedures, and the supernatants were collected in the same beakers for each sample. All of the supernatants were then extracted by the same SPE protocol with liquid samples.

To analyze the targeted pharmaceuticals, an ultra-high-performance liquid chromatography with tandem mass spectrometry (UHPLC-MS/MS), Q Exactive™ Focus MS, UltiMate™ 3000 RSLC System (Thermo Scientific, USA) equipped with Orbitrap high resolution, and a heated electrospray ionization source (H-ESI II) was conducted for analysis of targeted compounds. For the mobile phase, 0.1 % formic acid in Milli-Q water (solvent A) and 0.1 % formic acid in methanol (solvent B) were used. The gradient elution program with a flowrate of 0.2 mL/min was initiated with 20 % solvent B for 1 min with the solvent concentration then increased linearly to 90 % over 4 min. The solvent concentration was then held at 90 % for 2 min, before being reduced to 20 % for 3 min to equilibrate. A Thermo Hypersil GOLD™ C18 column (150 mm × 2.1 mm, 5 μm particle size, 175 Å pore size) was used at a constant temperature of 40 °C. The injection volume was 5 μL. The polarity was operated in the positive mode of ESI for MS parameters. The HESI source parameters consisted of an Aux gas heater temperature of 300 °C, capillary temperature of 250 °C, S-len RF level of 50 and spray voltage of 2.8 kV. Ultrapure nitrogen was employed both as the source gas for ions and as the gas used for collisions. It was utilized in the following proportions: 40 au for sheath gas, 10 au for auxiliary gas, and 2 au for sweep gas. The Orbitrap was operated in parallel reaction monitoring (PRM) mode with 35,000 resolution targeted-MS^2. The mass range in the PRM was scanned at 50.0–280.0 *m/z*. Maximum injection time was automated, and the automatic gain control (AGC) target in the PRM was established at a value of 5×10^4 ions. Normalized collision energy (NCE) was stepped at 20 and 30 V for PRM, while an isolation window was set at 3.0 m/z. The acquisition state for instrument control was operated in Xcalibur, and TraceFinder was used for processing the data. The SPE recovery was validated by the differences of the peak areas of the target compounds in the mixed standards before SPE (solution A) and after SPE (solutions B): R (%) = (A/B) × 100. The SPE recovery of the SMX was 80.55 %, while for the other compounds, it was 100 %. The limit of detection (LOD) and limit of quantification (LOQ) were calculated as the standard concentrations in line with signal to noise ratios (S/N) of 3:1 and 10:1, respectively. The LC-MS/MS Orbitrap data are provided in Supplementary Data Table S2 and Fig. S6. The standard calibration curves of each compound are provided at 0.5–25 μg/L and 50–1000 μg/L with a high level of accuracy ($R^2 > 0.99$).

2.5. Mass balance analysis of targeted pharmaceuticals

The mass balance of the targeted pharmaceuticals in each C3, which refers to different redox conditions in each bioreactor, was calculated, taking into account the removal mechanisms of biodegradation, sorption in the biomass (suspended sludge/biofilm) and wasted sludges using Eq. (2):

$$M_{inf} = M_{S,MVB} + M_{S,MLSS-C3} + M_{b,C3} + M_{eff} + M_{wasted\ sludge} \quad (2)$$

where M_{inf} and M_{eff} are pharmaceutical loads in the influent and effluent through the experimental run (μg), respectively, $M_{S,MVB}$ and $M_{S,MLSS\text{-}C3}$ are pharmaceuticals removed by adsorption in biofilm on MVB and MLSS, respectively (μg), $M_{b,C3}$ is amounts of pharmaceutical removed by biodegradation in compartment 3 (μg) and $M_{wasted\ sludge}$ is pharmaceutical load of wasted sludge from compartment 3 through the experimental run (μg).

Regarding the calculation for sorption by the biofilm on MVB, sorption by MLSS in C3, and sorption in the wasted sludges, Eqs. (3), (4) and (5) were used, respectively:

$$M_{S,MVB} = M_{MVB} \times C_s \quad (3)$$

$$M_{S,MLSS-C3} = Q \times MLSS \times C_s \times t \quad (4)$$

$$M_{wasted\ sludge} = C_s \times MLSS \times Q_w \times t \quad (5)$$

where M_{MVB} is the mass of biofilms on the MVB in compartment 3 (g), C_S is the concentration of pharmaceuticals in biomass (μg/g.MLSS), Q is the flow rate of bioreactors (L/day), *MLSS* is mixed liquor suspended sludge (g.MLSS/L), t is time of the experimental run (day) and Qw is the wasting sludge flowrate (0.405 L/d).

Moreover, sludge water distribution coefficient (K_d) in C3 of each bioreactor was calculated using Eq. (6):

$$K_d = \frac{C_s}{C_e} \quad (6)$$

where K_d is the sludge water distribution coefficient (L/kg. MLSS), C_S is the concentration of pharmaceuticals in biomass (μg/kg. MLSS) and C_e is the concentration of pharmaceuticals in the aqueous phase (μg/L).

2.6. Batch test for degradation kinetics and adsorption isotherm

Batch biotransformation experiments were performed to observe and compare the kinetic degradation of the targeted pharmaceuticals under anaerobic and microaerobic conditions. High-rise building wastewater was used in the batch test, and it was spiked with the targeted pharmaceuticals in various concentrations in 500 mL beakers placed on a magnetic stirrer plate to ensure proper mixing. The MVBs (biofilms) in C3 from R1 (for the anaerobic batch test) and R2 (for the microaerobic batch test) were filled into the beaker at 30 % by volume, and the suspended sludges in C3 from each bioreactor were washed with Milli-Q water, centrifuged for 5 min, and diluted until the corresponding beakers contained 300 mg MLSS/L. Four concentration levels ranging from ten to thousands μg/L were used for this test. The methanol added from the spiking in highest concentration contributed to COD of approximately 700 mg/L in the batch system. To ensure the COD was consistent in all experiments, more methanol was added to the remaining reactor to achieve a consistent COD value of 700 mg/L. Before starting the batch test, the beakers were covered with aluminum foil to prevent photo degradation and sealed with parafilm in the anaerobic batch test to prevent the oxygen from entering into the reactors. For the microaerobic batch tests, air was supplied corresponding to DO at 0.3–0.7 mg/L using a DO controller throughout the test. The batch biotransformation test was conducted for 120 h, which is adequate for observing the short-term biodegradation of microbial sludges (van Bergen et al., 2021). Five sampling times were continuously applied at 0, 24, 48, 96 and 120 h for analysis of the remaining targeted pharmaceuticals. To conduct the control batch experiments under anaerobic and microaerobic conditions, virgin MVBs at 30 % by volume were mixed with high-rise building wastewater and pharmaceuticals at a concentration of 250 μg/L. The targeted pharmaceuticals were then analyzed at 0, 48 and 120 h to ensure that no removal of the pharmaceuticals occurred in the absence of biofilms and suspended sludge.

The kinetics for the biotransformation of the pharmaceuticals were represented by pseudo first order kinetics (Carneiro et al., 2019; Prasertkulsak et al., 2019; Svendsen et al., 2020) as shown in the following Eq. (7):

$$ln\left(\frac{C_t}{C_0}\right) = -k_{bio} \times C_0 \quad (7)$$

where C_t and C_0 represent the pharmaceutical concentrations in aqueous phase at the sampling times and initial time of biotransformation, respectively (μg/L), k_{bio} is the biodegradation constant (h^{-1}) and t is time (h).

Identical parallel batch experiments were conducted under microaerobic and anaerobic conditions to study the adsorption isotherms without biodegradation and determine which are related to each pharmaceutical adsorption behavior. Biofilms and suspended sludges were inactivated by adding 0.10 % (*w/v*) of sodium azide solution (NaN_3) to inhibit the biodegradation of the targeted pharmaceuticals, in accordance with what was recommended in the literature (Wang et al., 2023). Then, the pharmaceutical concentrations in the aqueous phase were measured at 0 h and steady state concentration (120*h*). The results obtained from the batch adsorption tests were fitted to the Langmuir isotherm model given by Eq. (8) and the Freundlich isotherm model represented by Eq. (9) as described below:

$$\frac{1}{q_e} = \frac{1}{K_L \times q_{max}} \times \frac{1}{C_e} + \frac{1}{q_{max}} \quad (8)$$

$$logq_e = \log K_f + \frac{1}{n} \times \log C_e \quad (9)$$

where q_e is the equilibrium concentration of the targeted compounds adsorbed in the biomass (μg/g), K_L is the Langmuir coefficient (L/μg), q_{max} is the maximum adsorption capacity of the targeted compounds onto the biomass (μg/g), C_e is the equilibrium concentration of the targeted compounds in the aqueous phase (μg/L), K_f is the Freundlich coefficient and $\frac{1}{n}$ represents the reciprocal of the adsorption intensity.

2.7. Microbial community analysis

The biofilms on the MVBs in C3 of each bioreactor (MVB-R1, MVB-R2 and MVB-R3), the suspended sludges in C3 of each bioreactor (SS C3-R1, SS C3-R2 and SS C3-R3) and the fixed film samples in C1 of R2 (FF1-R2) and C2 of R2 and R3 (FF2-R2 and FF2-R3) at the end of the experimental run were gathered and preserved at −20 °C until they were ready for analysis. The DNA extraction process was carried out using a DNeasy PowerSoil Pro DNA Kit (Qiagen, USA). The total genomic DNA of all samples was extracted using DNeasy PowerSoil Pro DNA Kit (Qiagen, USA). The primers of 341F (TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGCCTACG GGNGGC WGCAG) and 805R (GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGGA CTACHVGGG TATCTAATCC and 2× sparQ HiFi PCR master Mix (QuantaBio, USA) were used to amplify the 16S rRNA gene, targeting the V3-V4 regions of bacteria and archaea. The PCR amplification began with an initial denaturation step at 98 °C for 2 mins followed by 30 cycles, each consisting of denaturation at 98 °C for 20 s, annealing at 60 °C for 30 s, and extension at 72 °C for 1 min. Finally, the single step of the final extension was performed at 72 °C for 1 min. Following the PCR amplification, purification of the 16S amplicons was carried out using sparQ Puremag Beads (QuantaBio, USA). Each sample was combined with 5 μL of Nextera XT index primer in a 50 μL PCR reaction, and the amplification was performed for 8–10 additional cycles using the mentioned PCR conditions. Bioinformatics analysis was conducted using QIIME2 version 2020.8 (Bolyen et al., 2019). The raw sequence data underwent quality filtering and demultiplexing using the q2-demux plugin. Subsequently, denoising of the sequences was performed using DADA2, implemented through the q2-dada2 plugin (Janssen et al., 2018). The beta diversity metrics of Principal Coordinate Analysis (PCoA) using weighted UniFracs were estimated through q2-diversity after subsampling, without replacement, to a total of 16,682 sequences (Faith, 1992).

2.8. Statistical analysis

Statistical analysis and fitting of the data to linear regression were performed using Origin lab ® and Microsoft Excel®.

3. Results and discussion

3.1. Effect of microaeration and sludge recirculation on biomass characteristics, VFAs and dissolved methane removal

Fig. 2(a) shows the ratio and the biomass concentrations of the biofilms on the MVBs and MLSS in C3 of the three bioreactors. It was found that the average biomass concentrations were significantly increased in the microaerobic AnBB-MBRs (R2 and R3) in both biofilms and MLSS as compared to R1: AnBB-MBR. The ratios of the biofilm/MLSS were reduced from 0.92 in R1 to 0.82 in R2, which indicates that microaeration played a vital role in the enhancement of suspended biomass growth rather than biofilm formation. However, the decline in MLSS of C3 in R3 as compared to R2 might be from the sludge recirculation to the entrapment of MLSS within a fixed film. In the presence of oxygen, microorganisms degrade organics and nutrients as the growth substrates which can provide higher growth yields (biofilm and suspended biomass) than in anaerobic conditions (Thoai et al., 2023). In addition, a small amount of oxygen 0.93 L_{O2}/L_{feed} could help increase the microorganism activities, promoting the formation of biomass, resulting in a broader and larger range of particle size distribution (Fig. 2 (b)) compared to the sole anaerobic process. The particle size distribution in the R2 and R3 were in the range of 2 to 586 μm and 2 to 400 μm, respectively. On the other hand, in the anaerobic system (R1), the particle size distribution was in the range of 0.5 to 272 μm, indicating a narrower range of smaller sizes of particles in the reactor.

SEM analysis was conducted to explore the microbial morphology of virgin MBR surface and the biofilm on MVB in the C3 compartment of R1, R2 and R3 as shown in Fig. 2(c), (d), (e) and (f), respectively. It is clearly shown that microaeration and sludge recirculation led to the difference in the biofilm morphology in the MVBs. In the anaerobic biofilm of R1, the thinner layers of filamentous bacteria were forming loosely on the surface of the carrier. The reduction of the filamentous bacteria and high dense flocs from the bolder biofilm layer was obvious in R2. However, the incorporation of microaeration and sludge recirculation in R3 seems to have resulted in anaerobic filamentous bacteria on the biofilm of the MVB, which is in line with the result of microbial consortium in Section 3.5. These SEM results are also in accordance with the biofilm concentrations whereby the microaerobic biofilms (R2 and R3) had a greater mass than the anaerobic biofilm (R1), whereby biofilm growth was not constrained to the entire surface area of the MVB. Previous research found that the biofilm formation of Candida species in growth medium was greater formed under microaerobic conditions rather than under anaerobic conditions (Koshikawa et al., 2022).

The profiles of the removal performances of the three AnBB-MBR systems in terms of COD, VFA species and suspended sludges are shown in Supplementary Data Table S1. The results show that COD removal was enhanced to 82.5 % and 81.8 % in R2 and R3, respectively while 64.9 % of COD removal was observed in R1. Organic matters and nutrients were found to increase the cell population in both suspended biomass and biofilms under microaerobic conditions. Generally, aerobic bacteria has a higher growth yield compared to strictly anaerobic bacteria due to the efficient metabolic pathways and abilities of each bacterial type to adapt and utilize various food sources (Chen et al., 2020). Acetic acid was the major VFA in the permeate of R1 (11.16 mg/L), and all species of VFAs were found in the permeate of R1. Microaerobic AnBB-MBRs (R2 and R3) could also enhance acetic acid removal to 100

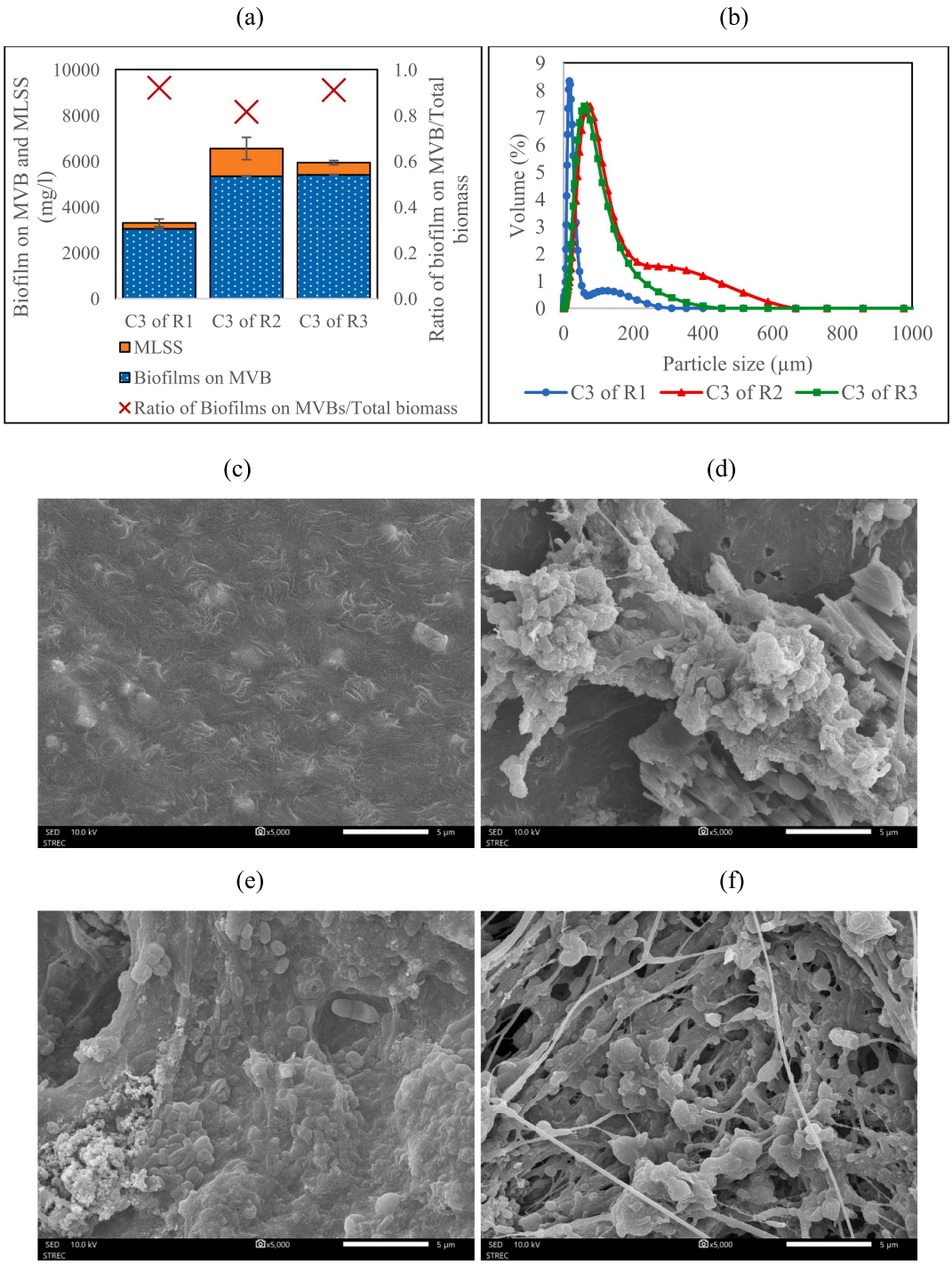


Fig. 2. The ratios and concentrations of Biofilms on MVBs and MLSS in C3 (a); particle size distribution of biomass in C3 (b); and surface morphology of biofilms by SEM (×5000) of MVB in C3 for virgin MVB (c), R1 (d), R2 (e) and R3 (f).

% and 96.9 % in R2 and R3, respectively when compared to R1, which was at 67.6 %, in the same way as the removal of other VFA species. The higher reduction of acetic acid and other VFA species under micro-aerobic conditions (C3 of R2 and R3) might be associated with the utilization as carbon sources and electron donors in the heterotrophic denitrification and cell synthesis processes (Fan et al., 2022).

Dissolved methane was found in lower concentrations in the permeate of R2 and R3 than in that of R1 by up to 87.94 % and 100 % (23.38 ± 0.27, 2.82 ± 0.03 and 0 mg/L in R1, R2 and R3, respectively) as shown in Supplementary data Fig. S2. The addition of oxygen has the potential to mitigate the effects of dissolved methane in anaerobic effluents. For example, Hatamoto et al. (2010) employed an encapsulated biological down-flow hanging sponge reactor at HRT 2 h with air flow of 3.8 $m^3.m^{-3}.d^{-1}$ which achieved dissolved methane removal up to 97 %. Cabral et al. (2020) achieved a dissolved methane reduction of up to 55 % in a microaerobic bioreactor with airflow of 240 $L.m^{-3}.d^{-1}$,while a 33 % reduction of dissolved methane was observed in a none aerated-bioreactor. Silva-Teira et al. (2017) treated the effluent of UASB with the anoxic/aerobic MBR, which resulted in up to 80 % removal of the dissolved methane. In this study, the AnBB-MBR system could achieve

an 87.94–100 % reduction of dissolved methane in the treated effluent, when the microaeration and microaeration with sludge recirculation were applied at 0.1 L_{air}/min equal to 0.93 O_2/L_{feed}. Moreover, Methylosarcina, Methylocaldum and Methylocystis, known as methane oxidizing bacteria or methylotrophs, which contribute to the oxidation of dissolved methane were also found in C3 of R2 and R3 at a relative abundance higher than that detected in C3 of R1, which will be discussed later in Section 3.5. Therefore, these microorganisms in the microaerobic systems could help oxidize the dissolved methane as a carbon source to prevent the release of this potential greenhouse gas into the atmosphere.

3.2. Effect of microaeration and sludge recirculation on the fate and removal characteristics of pharmaceuticals in the AnBB-MBR system

Fig. 3(a), (b), (c) and (d) show the influent and permeate concentrations of CIP, CAF, SMX and DCF, respectively over 115 operation days, while the profiles of each compartment and the average removal efficiencies of the three AnBB-MBR systems are shown in Fig. 4(a).

It can be seen that both microaeration (R2) and microaeration with sludge recirculation (R3) had a positive impact on the removal efficiencies for most of the targeted compounds in the AnBB-MBR. The highest removal of CIP could be achieved by microaeration at 78.0 % in R2, compared to 63.3 % in R1 without microaeration and 60.2 % in R3 with microaeration and sludge recirculation. >90 % of CAF was

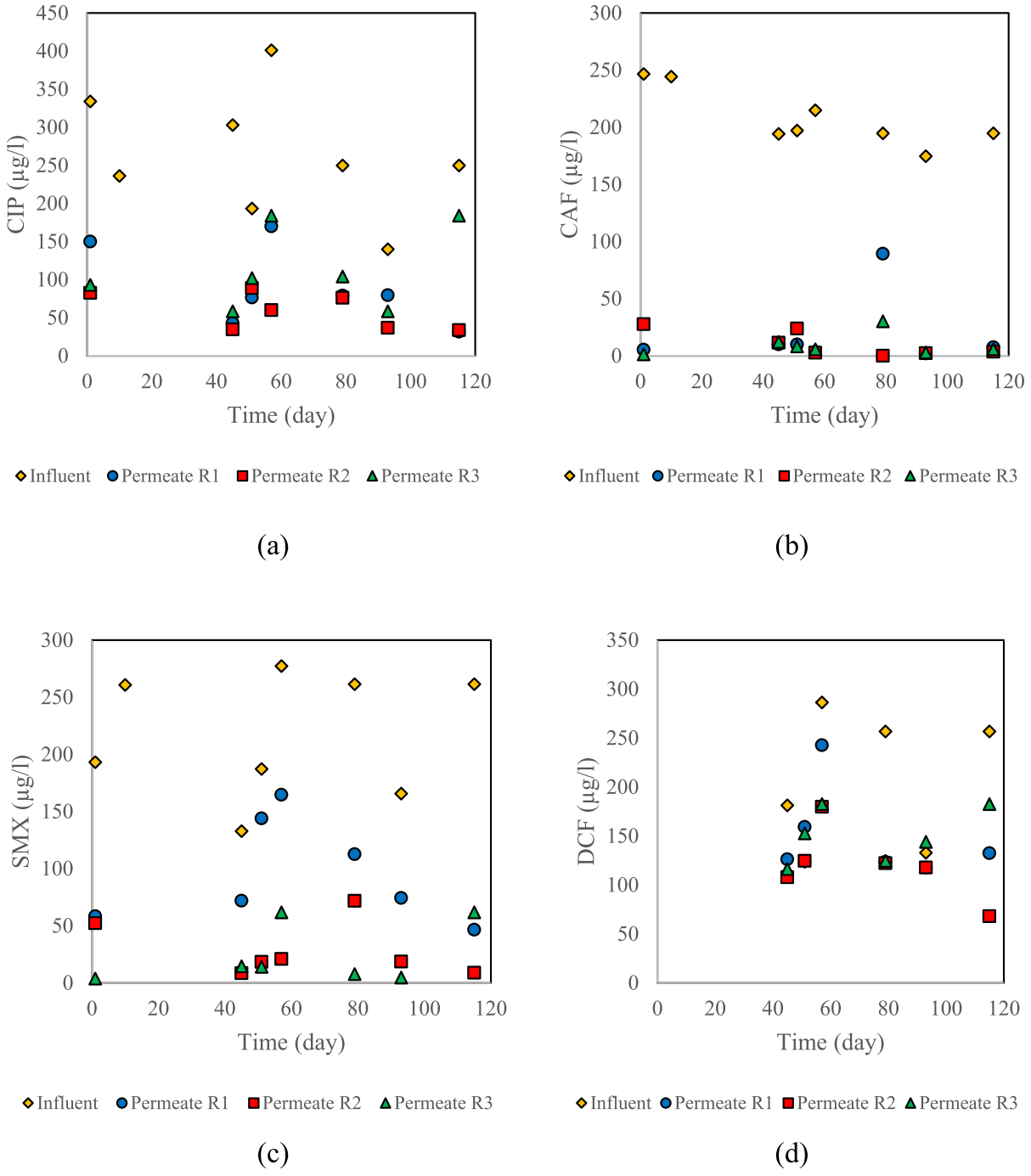
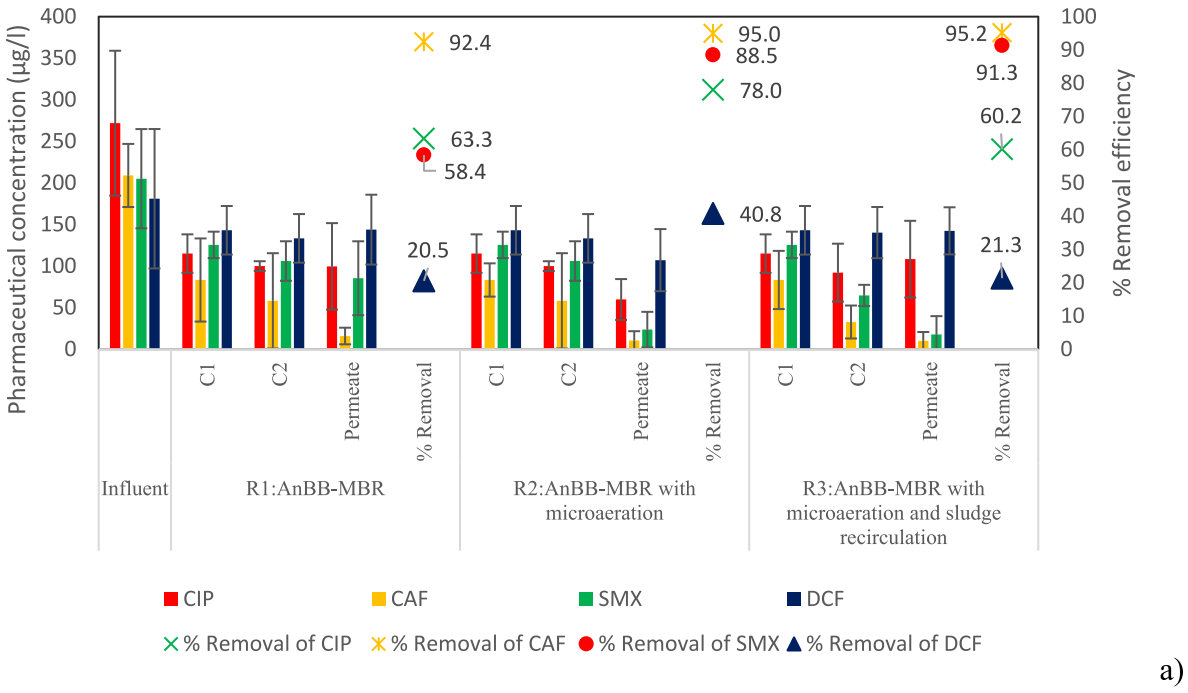
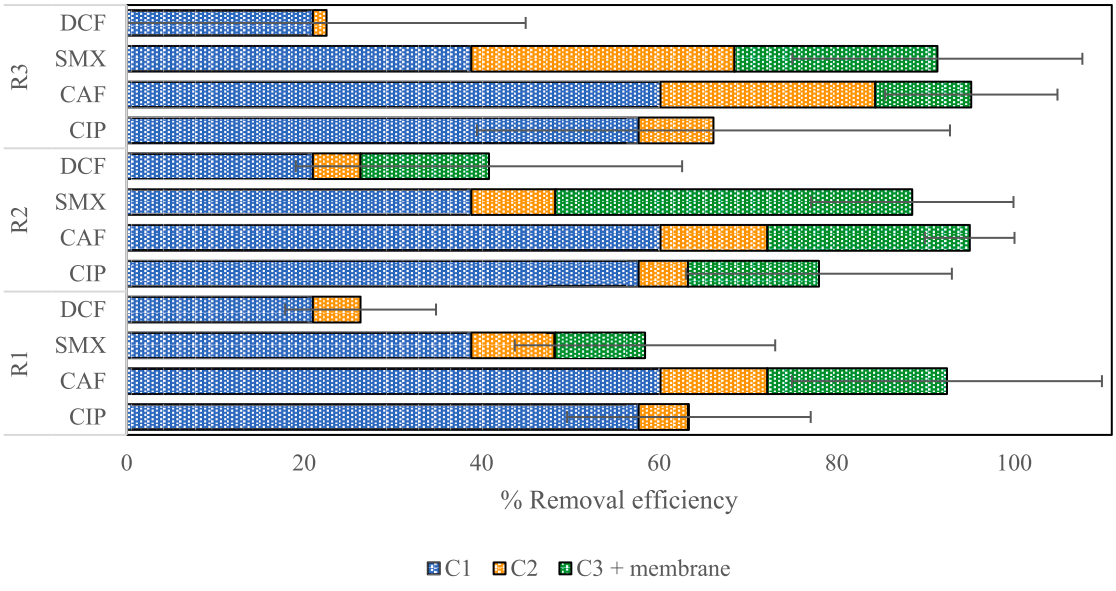


Fig. 3. Influent and permeate concentrations of CIP (a), CAF (b), SMX (c), and DCF (d) during the experimental run 115 days of three bioreactors.



a)



b)

Fig. 4. (a) Concentration profiles and the removal efficiencies of the target compounds in each compartment; (b) Comparing the fraction of average removal of the targeted compounds in each compartment of three bioreactors (Number of samples, n = 8).

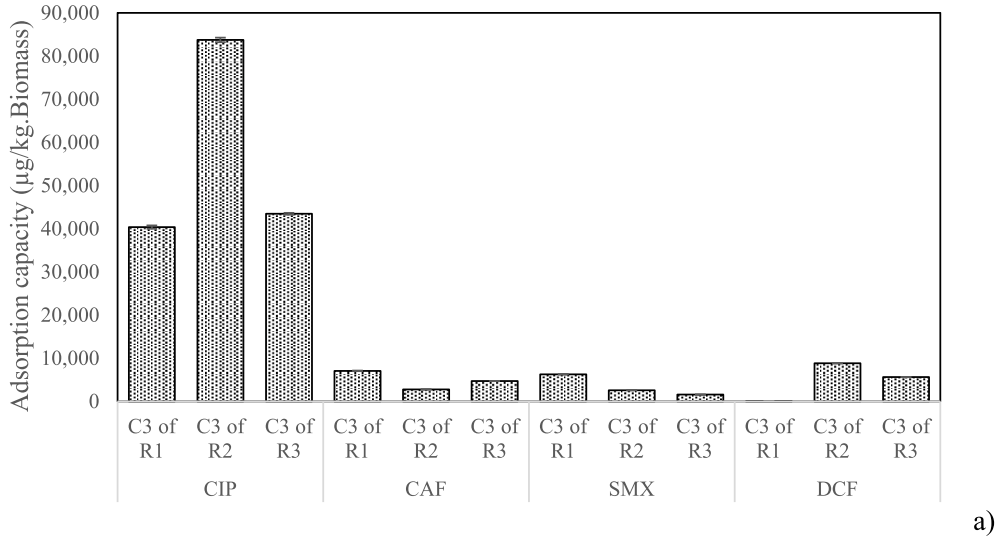
removed efficiently regardless of bioreactor conditions. Park et al. (2017) and Wijekoon et al. (2015) also reported that CAF is an easily biodegradable compound in various redox environments. Enhancing the removal of SMX could be significantly achieved with R2 (88.5 %) and R3 (91.3 %) rather than R1 (58.4 %). However, the removal of DCF was only enhanced in R2 (40.8 %), compared to R3 (21.3 %) and R1 (20.5 %). Poor removal of DCF was observed in several WWTPs and conventional MBRs since it is one of the refractory compounds by biological treatment (Luo et al., 2015; Monsalvo et al., 2014). Bacteria living in the presence of oxygen could quickly release enzymes like hydroxylase and oxygenase, which facilitate the breakdown of pharmaceutical substances through a process involving electrophilic reactions, and this is especially effective when dealing with compounds that have functional groups donating electrons (Harb et al., 2016; Liu et al., 2020).

Comparing the average removal of targeted compounds in each compartment as illustrated in Fig. 4(b), it seems that the fixed film compartment (C1) of the AnBB-MBR system achieved the highest removal percentages for all compounds. The microbial community in C1 demonstrated adaptability to elevated levels of pharmaceutical loads, thereby fostering pharmaceutical degradation. Subsequently, as the

concentration of pharmaceuticals diminished, encompassing the microbial consortia even under the same anaerobic conditions in C2 of both R1 and R2, the pharmaceutical removal efficiencies were reduced. In the case of C2 of R3, the observed highest percentage of pharmaceutical removal in comparison to C2 of other reactors was attributed to the implementation of sludge recirculation, which induced anoxic/microaerobic conditions, thereby fostering a diverse microbial community capable of enhancing pharmaceutical degradation and creating an environment conducive to oxygen availability. It is noteworthy that C3 of R2 exhibited the most substantial contribution to pharmaceutical removal compared to C3 in both R1 and R3. Consequently, it can be inferred that the utilization of single microaeration proves to be the most effective strategy for enhancing pharmaceutical removal efficiency.

The sorption of compounds can be influenced by various characteristics of the biomass. Biomass's adsorption capacity for each compound in C3 is shown in Fig. 5(a). CIP is the only one of the compounds that is highly adsorptive onto biomass with concentrations higher than 10,000 μg/kg.biomass, and it was most significantly absorbed by the microaerobic biomass in C3 of R2 compared to the biomass from the other reactors. The average K_d and log K_d values as shown in Supplementary data Table S3 can be used to classify the compounds as highly sorptive (log $K_d > 3$), moderately sorptive ($2 <$ log $K_d < 3$) and lowly sorptive (log $K_d < 2$) compounds (Lakshminarasimman et al., 2018). CIP was the compound adsorbed the most by the microaerobic R2 biomass (C3 of R2 with log K_d is 3.09), while it was more moderately adsorbed by the anaerobic biomass in C3 of R1 and the microaerobic recirculated biomass in C3 of R3. CAF was moderately adsorbed under both conditions with K_d values of 380.8–503.2 L/kg.MLSS in similar levels to the anaerobic/anoxic/aerobic–MBR sludges, which had K_d values in the range of 300–800 L/kg.MLSS (Xue et al., 2010).

The adsorption of SMX by the microaerobic biomass was moderate (log $K_d = 2.12$), whereas low adsorption (log $K_d = 1.85$) of SMX occurred



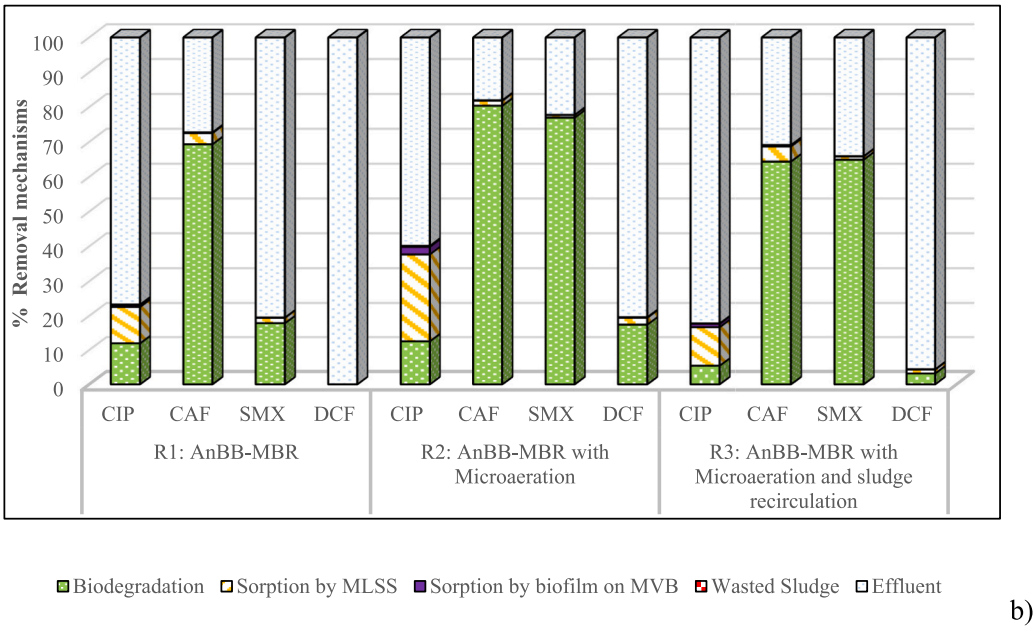


Fig. 5. a) Adsorption capacity of targeted pharmaceuticals in biomass in C3 compartment in each bioreactor b) Mass balance of selected pharmaceuticals in C3 of three bioreactors.

in the anaerobic biomass. At pH > 5.8, SMX (pKa2 = 5.7) exists in an anionic form (Avisar et al., 2010), which is likely to result in minimal adsorption by negatively charged biomass. A previous study by Lakshminarasimman et al. (2018) discovered that SMX exhibited a moderate level of adsorption that was consistent across different redox environments, including anaerobic, anoxic, and aerobic sludges, in biological nutrient removal treatment systems. However, other comparative studies have observed that aerobic sludges adsorbed SMX much more than anaerobic sludges did (Alvarino et al., 2014; Iliopoulou et al., 2023), which concurs with the results from this study. For DCF, it was uniformly reported to exhibit very low sorption regardless of the redox conditions of the microbial sludges (Iliopoulou et al., 2023; Song et al., 2018; Wijekoon et al., 2015). This study can confirm that the sorption of DCF was very low ($\log K_d < 2$) with no significant difference between the sorption by the microaerobic and the anaerobic biomass.

Considering the fate and removal mechanisms of the targeted compounds of the three bioreactors, mass balance analysis was conducted at C3 in terms of biodegradation, adsorption by MLSS and biofilm on the MVB, excess sludge discharge and effluent, as shown in Fig. 5(b). Regarding the molecular structures and function groups of the targeted pharmaceuticals, it is evident from the result of this study that microaeration in C3 of R2 is the most likely to have a positive effect on the removal mechanisms by enhancing the adsorption of CIP and the biodegradation of CAF, SMX and DCF. Sorption was the main removal mechanism of CIP in C3 of R2, but its effectiveness was reduced in the biomass in C3 of R1. In R3, the adhesion of CIP to MLSS might be likely to experience detachment due to the use of sludge recirculation and changes in the floc conditions during the transition between anaerobic and microaerobic phases, while the suspended sludge in R2 is in a microaerobic state. These results concur with the literature on CIP removal in aerobic MBRs, whereby CIP was entirely removed by adsorption onto the sludge in the aerobic MBR pilot plant (Gurung et al., 2019; Kim et al., 2014).

The piperazinyl-positive charge, amine positive charge and negative charge of the carboxyl group in the structure of CIP, which exhibit the zwitterion ions in the neutral pH (around pH = 7), could induce electrostatic interaction with negative charged biomass, while divalent ions (Ca^{2+} and Mg^{2+}), which were generally bound on to negatively charged sludge, can also induce anionic carboxyl to form cation bridging and cation exchange with protonated amine in zwitterionic CIP form (Polesel et al., 2015) leading to the adsorption of CIP onto the negatively charged biomass by divalent bridges. In this research, the average zeta potentials of the biomass in C3 were − 22.5 mV, −19.9 mV and − 21.5 mV in R1, R2 and R3, respectively (Supplementary Data Table S3), which implies that the micro-aerated sludges with lesser negative charges probably tended to be more suitable for the adsorption of the zwitterion ions of CIP than the anaerobic sludges. Higher biomass concentrations and a broader range of sludge particle sizes were previously reported as enhancing the adsorption capacity of micropollutants in wastewater (Alvarino et al., 2014; Dorival-Garcia et al., 2013). Thus, higher MLSS concentrations in C3 of R2 might be a reason for the higher adsorption of CIP in C3 of R2 than in the other bioreactors due to the microaerobic biomass having more specific surface area and a broader range of particle sizes, thereby providing various adsorption sites with multilayers. The broader ranges of particle sizes in aerobic MBR have been previously reported as being beneficial to its capacity to adsorb micropollutants due to it having a greater surface area compared to anaerobic biomass. (Monsalvo et al., 2014).

More than 90 % of CAF was efficiently removed by biodegradation regardless of the reactor configurations or redox environment. CAF is hydrophilic molecules and contains the methyl group (-CH_3), which is an electron donating group that can be easily degraded by demethylation processes from aerobes and anaerobes (Teoh et al., 2022) and oxidation from aerobes (Gummadi et al., 2012). For SMX, previous studies have consistently found that it was removed by biodegradation, while low adsorption was observed in various redox conditions the results of this study (aerobic, anaerobic, anoxic/aerobic) and wastewater treatment technologies (Alvarino et al., 2016; Alvarino et al., 2014; Iliopoulou et al., 2023; Wijekoon et al., 2015). In line with this study, biodegradation is the major pathway for the removal of SMX, and it has been found that the use of microaeration (0.93 L_{O2}/L_{feed}) and microaeration with sludge recirculation can enhance the biodegradation efficiency. The presence of electron-donating groups, amine group (−NH_2) and -CH_3, in SMX structures might promote its biodegradation in oxygen availability. Stadler et al. (2015) also found that 62 % of SMX was biodegraded in the microaerobic process and by over 35 % and 38 % in anoxic/aerobic and aerobic sequencing batch reactors, respectively. The reductive biotransformation of SMX under less negative redox potential or in oxygen availability as indicated as ORP could facilitate the cleavage of the sulfonyl group and isoxazole ring (Alvarino et al., 2018).

The removal efficiencies of DCF via biodegradation were 17.5 % in C3 of R2, >5 % in C3 of R3, and no removal observed in C3 of R1. DCF, an aromatic amine, exhibits the propensity to generate nitro derivatives in the presence of nitrites, and these derivatives tend to revert to the original compound upon the reduction of nitrites in an anaerobic environment (Iliopoulou et al., 2023). The removal pathways of DCF followed the same trends as observed in previous comparative studies of the aerobic and anaerobic process degradation of DCF, which shows its low biodegradability and insignificant difference in removal outcomes under these redox conditions (Alvarino et al., 2018; Alvarino et al., 2014; Iliopoulou et al., 2023; Liu et al., 2020). The presence of the chloro group (—Cl), which is an electron withdrawing group might render diclofenac recalcitrant for biodegradation processes. Dechlorination was reported as the main mechanism in the removal of DCF under anaerobic reductive condition by using chlorine as the final electron acceptors by anaerobic microbes (Yang et al., 2022). Moreover, Poirier-Larabie et al. (2016) stated that oxygen played a vital role in the hydroxylation of the phenyl of DCF.

Table 1 depicts some previous anaerobic MBR technologies in a variety of reactor configurations, wastewater types and concentrations and operating conditions for the removal of pharmaceuticals such as CIP, CAF, SMX and DCF, compared to the newly developed AnBB-MBR systems in this study. It is shown that the AnBB-MBR systems operated at relatively higher flux rates (12.6 $L.m^{-2}.h^{-1}$) compared to most of the previous anaerobic MBRs. Nevertheless, the efficacy of pharmaceutical removal in AnBB-MBRs is significantly superior to that reported in other studies as shown in Table 1. This indicates that AnBB-MBRs is an efficient wastewater treatment approach for removal of pharmaceuticals. Moreover, it can be concluded that microaeration at 0.1 L_{air}/min (0.93 L_{O2}/L_{feed}) enhanced the removal of CIP from 63.3 % to 78 %, CAF from 92.4 % to 95 %, SMX from 58.4 % to 88.5 % and DCF from 20.5 % to 40.5 % as compared to AnBB-MBR without microaeration (R1).

3.3. Effect of microaeration and sludge recirculation on membrane fouling reduction in the AnBB-MBR system

The increment of TMP, as shown in Fig. 6(a), is used to indicate the rate of membrane fouling in the AnBB-MBR systems. Over 115 days of operation, the membrane of R3 and R2 had fouled only 6 and 7 times, respectively. Membrane fouling was lower than that of R1 which fouled 15 times. The average fouling index of R3 and R2 were 21.4 ± 11.2 and 21.7 ± 8.9 mmHg/d with the specific fouling rates of 6.16 and 6.26 $mmHg/L_{permeate}/m^2$, respectively, which were lower than that of R1 by approximately 65 % (see supplementary materials Fig. S3). Evidently, the incorporation of microaeration significantly improved the effectiveness of membrane fouling reduction enabling the prolonged operating period of the AnBB-MBR system, while the integration of microaeration with sludge recirculation (R3) could slightly enhance membrane filterability in AnBB-MBR as compared to the single microaerated AnBB-MBR (R2). The wide range of particle sizes (Fig. 2 (b)) in R2 and R3, they adhered to the membrane surface, could slow the fouling because of the formation of voids or gaps between particles with

Table 1
Comparation of the removal of targeted pharmaceuticals between AnBB-MBR systems and recent advanced AnMBR technologies.

AnMBR configurations	Wastewater characteristics	Operating conditions			Pharmaceutical concentrations	Pharmaceutical removal efficiency (%)	Reference
		Flux (L.m^2.h^{-1})	HRT (h)	pH			
Submerged AnMBR	Synthetic municipal wastewater, COD = 500 mg/L	2.67–5.33	6–12	7.4	CIP = 0.5–4.7 mg/L	CIP = 20–76.4	(Do et al., 2022)
Submerged upflow AnMBR	Synthetic swine wastewater, COD = 3000 mg/L	2	22	7.5	SMX = 100 μg/L	SMX = 81.3–91.7	(Cheng et al., 2021)
External ceramic AnMBR	Synthetic wastewater, TOC = 2152.9 mg/L	0.94	48	6.9	DCF = 5 μg/L	DCF = 14	(Liu et al., 2020)
Side stream PVDF AnMBR	Synthetic municipal wastewater, COD = 810–3330 mg/L	3	24	7.0	SMX = 10 mg/L	SMX = 97.2	(Wei et al., 2019)
Staged fluidized AnMBR	Real municipal wastewater, COD = 115.49 mg/L	1.34	7.5	7.0	DCF = 1.6–3.0 μg/L	DCF ≤0	(Chen et al., 2019)
submerged AnMBR	Synthetic coffee processing wastewater, COD = 42,400 mg/L	0.43	120	4.8	CAF = 470 mg/L	CAF = 87.5	(Chen et al., 2018)
AnBB-MBR (R1)	Real building wastewater, COD = 245.8 mg/L	12.6	30	7.4	CIP, CAF, SMX and DCF = 250 μg /L	CIP = 63.3 CAF = 92.4 SMX = 58.4 DCF = 20.5	This study
AnBB-MBR with microaeration (R2)	Real building wastewater, COD = 245.8 mg/L	12.6	30	7.4	CIP, CAF, SMX and DCF = 250 μg /L	CIP = 78 CAF = 95 SMX = 88.5 DCF = 40.8	This study

significantly different sizes. In contrast, the particles in R1 were small with a narrow size range, which resulted in intimate accumulation on the membrane surface and pore blockage, potentially causing irreversible fouling (Vela et al., 2009). Yao et al. (2021) found that the biocake which formed during the fouling process of AnMBR exhibited a consistently smooth surface and a compact structure, while the biocake layer in the aerobic MBR was thicker and exhibited rougher and more porous properties.

Fig. 6(b) and (c) show the fouling components (proteins and carbohydrates) from SMP, L-EPS and T-EPS in the suspended biomass and membrane foulants, respectively. It was found that the proteins in SMP were the most commonly occurring component in the suspended biomass causing more clogging on the membrane of R1, contributing to be the foulant components in the R1 rather than those in R2 and R3. This finding concurs with previous study, which reported that the majority of the foulant in the AnMBR originated from the retained proteins and SMP fractions, which caused the membrane fouling in the AnMBR; however, the SMP in the aerobic MBR was capable of traversing the membrane pores and was detected in the permeate (Xiong et al., 2016). The concentration of SMP was generally greater in the sludges of AnMBR compared to the aerobic MBR (Wang et al., 2018; Yurtsever et al., 2015). This higher SMP content in AnMBR contributed to the more rapid development of severe membrane fouling in this system.

Additionally, certain types of bacteria have been found to have a negative impact on membrane fouling. Some intracellular L-EPS were released by anaerobic filamentous bacterial autolysis, and some biodegradable L-EPS could be preferentially consumed by anaerobic filamentous bacteria as a growth substrate (Hu et al., 2022). Thus, higher levels of L-EPS in the anaerobic R1 than in R2 might be associated with Pelolinea, Caldisericum, Longilinea and Leptolinea (as discussed in Section 3.5), which are filamentous bulb anaerobic bacteria that might cause severe membrane fouling (Mori et al., 2009). Desulfovibrio also likely influenced the accumulation of organic foulants, thereby affecting the behavior and development of biocakes in the AnMBR system (Yao et al., 2021). In addition, high-rate aeration has been found to be more effective in reducing membrane fouling compared to low-rate aeration due to the increased shear stress and air scouring effect, which help decrease the accumulation of substances that contribute to the membrane fouling (Belli et al., 2021). This might be associated with the lower fouling rate from the air scouring effect of the microaeration at 0.1l_{air}/min in R2 and R3.

Moreover, 3D-FEEM analysis was also conducted to identify the fluorescence characteristics of the organic substances and foulants in the influent and supernatants in C3 (MVB compartment). Fluorescent fractions were separated into four distinct regions (Coble, 1996) as outlined in our results as shown in Fig. 6 (d), (e), (f) and (g) for the influent, and mixed liquors in C3 of R1, R2 and R3, respectively. Peak A (EX: 220–250/EM: 330–380), which is considered to be tryptophan protein-like substances; Peak B (EX: 250–230/EM: 300–350), which represents soluble microbial byproduct-like substances; Peak C (EX: 300–350/EM: 400–450) and Peak D (EX: 250–260/EM: 380–460), which corresponds to humic acid-like compounds. The variation in fluorescence intensity can be associated with alterations in the levels of the compounds. The results show that all fluorescent peaks were decreased at position C3 in all bioreactors, when compared to the original building wastewater, which was rich in proteins, SMP, and humic acids. This indicates a significant reduction in organic fouling matters. The reduction of these compounds can be due to the presence of microbial populations within the AnBB-MBR system capturing and treating these large organic molecules present in wastewater in each compartment of the AnBB-MBR. When comparing the fluorescence intensity of the supernatants between C3 of each bioreactor, it was observed that the spectrum levels of Peak A and Peak B in R2 and R3 exhibited a decrease in fluorescence intensity as compared to R1. This suggests that employing microaeration is an efficient approach for diminishing protein and SMP components, which serve as the primary constituents of foulants in the AnBB-MBR system. This observation aligns with the findings related to the SMP and protein contents in the supernatant of suspended sludges within C3.

Furthermore, the functional groups of MBR foulants were analyzed by FTIR in the peak region between 1800 and 600 cm^{-1} (Schmitt and Flemming, 1998). As can be seen from Supplementary Data Fig. S4, the sharp peak in the range of 1047–1100 cm^{-1} was attributed to the stretching bond of C-O-C, indicative of substances resembling polysaccharides or carbohydrates, and the low peak intensity at 1380 cm^{-1} corresponds to the stretching vibration of C═O bonds in nucleic acids and polysaccharide groups (Li et al., 2022). The distinct peaks of around 1640 cm^{-1} (associated with the stretching of C═O and C—N bonds) and 1550 cm^{-1} (related to N—H deformation and C═N stretching) represent the presence of carbonyl groups, which are typical structural components found in secondary proteins, specifically within the amide I and amide II regions (Ratanatamskul and Kongwong, 2017). The findings exhibited a similar pattern for the functional groups between the

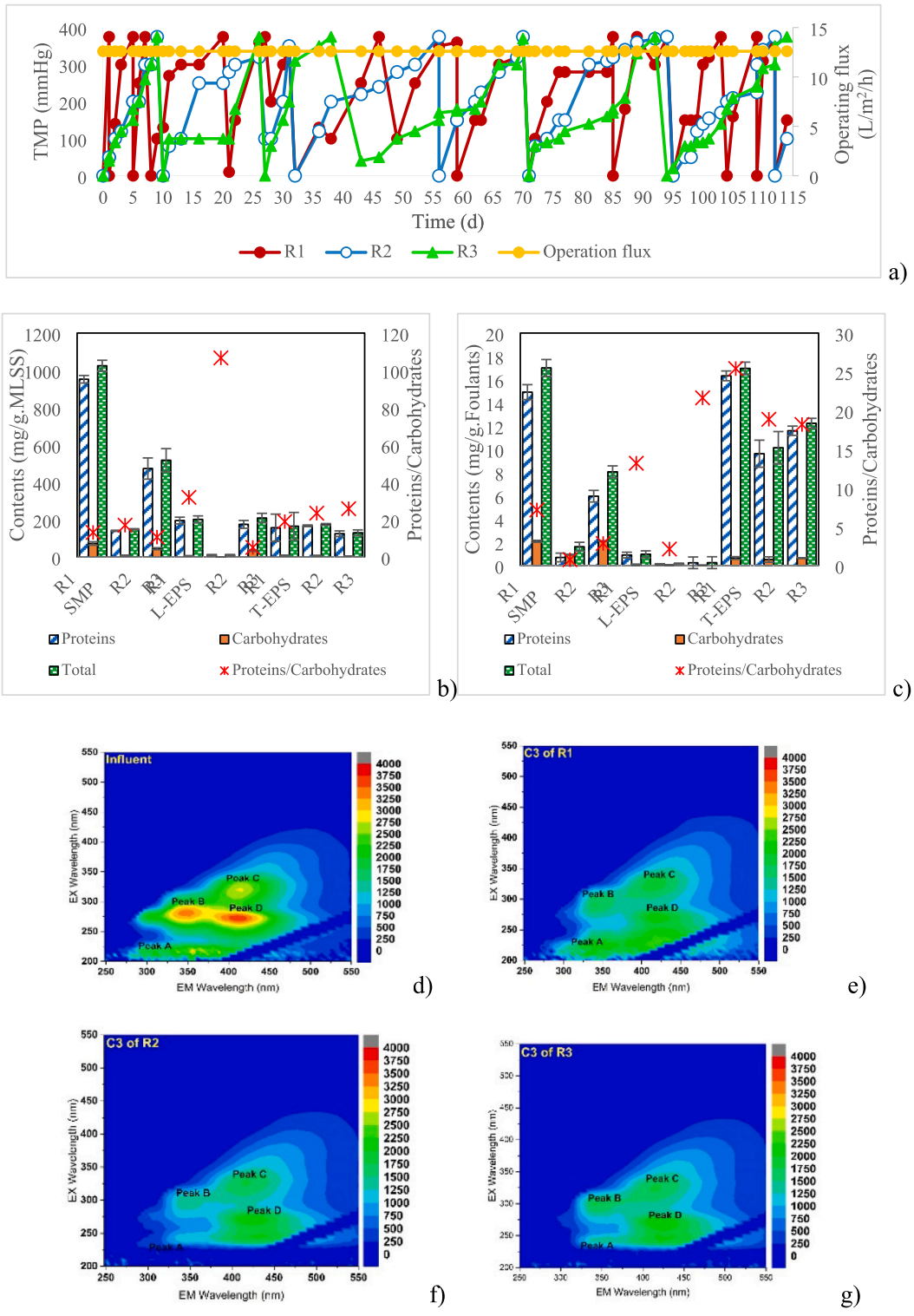


Fig. 6. Evaluation of TMP throughout the experiment run by maintaining the operating flux at 18.9 $L.m^{-2}.h^{-1}$ throughout the experiment run a); protein and carbohydrate contents in SMP, L-EPS, and T-EPS of suspended sludges b) and in membrane foulants c); FEEM analysis of the influent d); and mixed liquors in the C3 of R1 e), in C3 of R2 f) and in C3 of R3 g).

anaerobic and microaerobic foulants; nevertheless, variations in the absorbance intensity were detected between the three bioreactor systems. It was observed that the fouling intensity in R1 was the highest among all the compounds, suggesting that the fouling components of R1 were the most prominent factors contributing to a significant occurrence of membrane fouling. This finding aligns with the membrane foulant characteristics.

3.4. Effect of microaeration on biotransformation kinetics and adsorption isotherms of targeted pharmaceuticals

The biotransformation of the targeted pharmaceuticals over time in the batch tests is shown in Supplementary Data Fig. S5. It demonstrates that there was no removal of the targeted pharmaceuticals within a reaction period of 120 h under the control condition with no biomass in

the batch reactors. At the highest concentrations, all compounds totally inhibited biotransformation under anaerobic conditions, whereas only CAF and SMX inhibited biotransformation in microaerobic conditions. This might be from the lag phase of the biotransformation due to the adaptation of biofilms and sludge in C3 with relatively high pharmaceutical concentrations within 120 h of the batch test. In addition, it was found that CIP and CAF were not bio-transformed under anaerobic conditions at the lowest concentrations of 54.43 μg/L and 15.9 μg/L, respectively.

Table 2 shows the k_{bio} values of the targeted pharmaceuticals at various concentrations in the microaerobic and anaerobic batch degradation tests. The biotransformation kinetics under both conditions could fit to pseudo-first order kinetics with R^2 ranging from 0.77 to 0.99. The k_{bio} of CIP increased as the CIP concentration increased under microaerobic condition from 0.008 h^{-1} to 0.020 h^{-1}, and CIP did not inhibit the microaerobic microbial activity at the highest concentration (6801.92 μg/L), whereas the anaerobic biomass kept k_{bio} quite constant at 0.004–0.008 h^{-1}. In this study, it was observed that the ciprofloxacin in the building wastewater tended to undergo adsorption onto sludge rather than biodegradation in the presence of oxygen. This can explain why no inhibition of CIP was observed under microaerobic conditions at the highest concentration. Owing to the occurrence of both biodegradation and adsorption in the batch test, the higher k_{bio} values of CIP under microaerobic conditions than those under anaerobic conditions in all concentrations is partly due to the greater adsorption capacity by electrostatic interaction and cation bridging of the microaerobic biomass as described in Section 3.2. CAF is well known to be a highly biodegraded compound, regardless of redox conditions (aerobic, anoxic, and anaerobic) (de Wilt et al., 2018) providing the highest k_{bio} of 0.044 h^{-1} at the CAF concentration of 177.8 μg/L under microaerobic conditions.

This study also found that the k_{bio} values of SMX at relevant concentration in the environment at 28.55–291.46 μg/L were higher under microaerobic conditions (0.019–0.040 h^{-1}) than those under anaerobic conditions (0.005–0.009 h^{-1}). Likewise regarding the results for the removal and fate of SMX in Fig. 5(b), most previous research has found that SMX was more efficiently biotransformed in soils and sludges under the presence of oxygen with higher k_{bio} values than in an absence of oxygen (Biel-Maeso et al., 2019; Lakshminarasimman et al., 2018; Liu et al., 2010). For DCF, the relevant k_{bio} values and higher concentrations in the AnBB-MBR system (67.24–2314.15 μg/L) tended to be higher under microaerobic conditions than under anaerobic conditions. Alvarino et al. (2014) proposed a k_{bio} of DCF in activated sludge of 0.1 L/g.MLSS-d, which was higher than that in UASB at 0.007 L/g.MLSS-d, which concurs with the findings of this study. However, Iliopoulou et al. (2023) found that no k_{bio} values of DCF were observed in batch tests carried out in both anaerobic and aerobic MBRs due to its refractory biotransformation. In summary, the overall results with this developed AnBB-MBR system show that most of the target compounds were removed in greater amounts under microaerobic conditions than anaerobic conditions corresponding to the continuous operation of AnBB-MBRs.

In order to eliminate the effects of biodegradation, the inactivated biomasses were used to study the adsorption isotherms of the targeted compounds. It was found that the adsorption isotherms of the targeted pharmaceuticals under microaerobic and anaerobic conditions could be better fitted with the Freundlich model than the Langmuir model in the range of the test concentrations of all compounds ($R^2 > 0.9$). The fitted results of the Freundlich model are shown in Fig. 7 and Supplementary Data Table S4. The biomass surface heterogeneity and multilayer adsorption are Freundlich model which used K_f to indicate the affinity of compound to biomass or biosorption capacity and 1/n to indicate adsorbed surface heterogeneity with high adsorption intensity when the 1/n values is nearly zero (Cao et al., 2019). Due to the naturally negative charges of biomass, the positively charged compounds tended to be adsorbed by electrostatic interaction much more effectively than by neutral compounds and negative substances, respectively (Kahle and Stamm, 2007).

Freundlich isotherms show that microaerobic biomass could improve CIP adsorption onto the biomass as compared to anaerobic biomass, with microaerobic biomass and anaerobic biomass having K_f and 1/n values of 6.50 L/g and 0.42, and 1.72 L/g and 0.57, respectively. CIP has the lowest 1/n values as compared to the other compounds, which indicates that CIP was strongly adsorbed onto the biomass, and the K_f values were in line with previous research of CIP adsorption onto activated sludges and aerobic granular sludges (Ferreira et al., 2016). This result is in accordance with the reasons for the adsorption of CIP as earlier discussed in Section 3.2, whereby microaerobic biomass tended to adsorb CIP to a greater degree than anaerobic biomass did. For CAF and SMX, the differences in the adsorption isotherms between the microaerobic and the anaerobic biomass were found to be insignificant. CAF exists in a cationic form, while SMX and DCF are anionic compounds in neutral pH (pH around 7), which led to higher sorption of CAF onto negative sludges than those of two compounds and in accordance with their K_d values (Table S2). SMX and DCF were less absorbed onto the negatively charged biomass with K_f in the range of 0.01–0.12 L/g.

Table 2
Pseudo-first order kinetic parameters of targeted pharmaceuticals in microaerobic and anaerobic conditions during 120 h of batch biotransformation experiments.

Targeted pollutants	Initial concentration (μg/l)	Microaerobic condition		Anaerobic condition	
		k_{bio} (h^{-1})	R^2	k_{bio} (h^{-1})	R^2
CIP	87.36	0.008 ± 0.001	0.87	0.004 ± 6.22 × 10^{-4}	0.94
	142.93	0.011 ± 0.004	0.80	0.008 ± 0.002	0.85
	609.07	0.015 ± 0.003	0.88	0.007 ± 7.51 × 10^{-4}	0.96
	6801.92	0.020 ± 0.001	0.98	n.b.	–
CAF	15.90	0.041 ± 0.008	0.89	n.b.	–
	177.80	0.044 ± 0.018	0.85	0.024 ± 0.007	0.80
	897.86	0.012 ± 0.004	0.80	0.002 ± 7.0 × 10^{-4}	0.83
	7703.49	n.b.	–	n.b.	–
SMX	28.55	0.019 ± 0.002	0.93	0.009 ± 0.001	0.90
	291.46	0.040 ± 0.008	0.87	0.005 ± 0.001	0.84
	1111.87	0.001 ± 2.60 × 10^{-4}	0.87	0.004 ± 7.01 × 10^{-4}	0.93
	9600.85	n.b.	–	n.b.	–
DCF	13.69	0.013 ± 0.002	0.94	0.040 ± 0.016	0.86
	67.24	0.025 ± 0.003	0.97	0.012 ± 0.002	0.93
	307.83	0.049 ± 0.027	0.80	0.049 ± 0.004	0.99
	2314.15	0.018 ± 0.002	0.92	0.002 ± 0.001	0.88

Notes: n.b. is non biotransformation.

3.5. Microbial community structure

To gain insights into the role of microbial communities in the three bioreactors, samples of suspended sludge and biofilm on the MVB from C3 of the three systems and fixed film samples in C1 of R2 and C2 of R2 and R3 were collected after they reached a stable operational phase. These samples were then subjected to high-throughput sequencing by

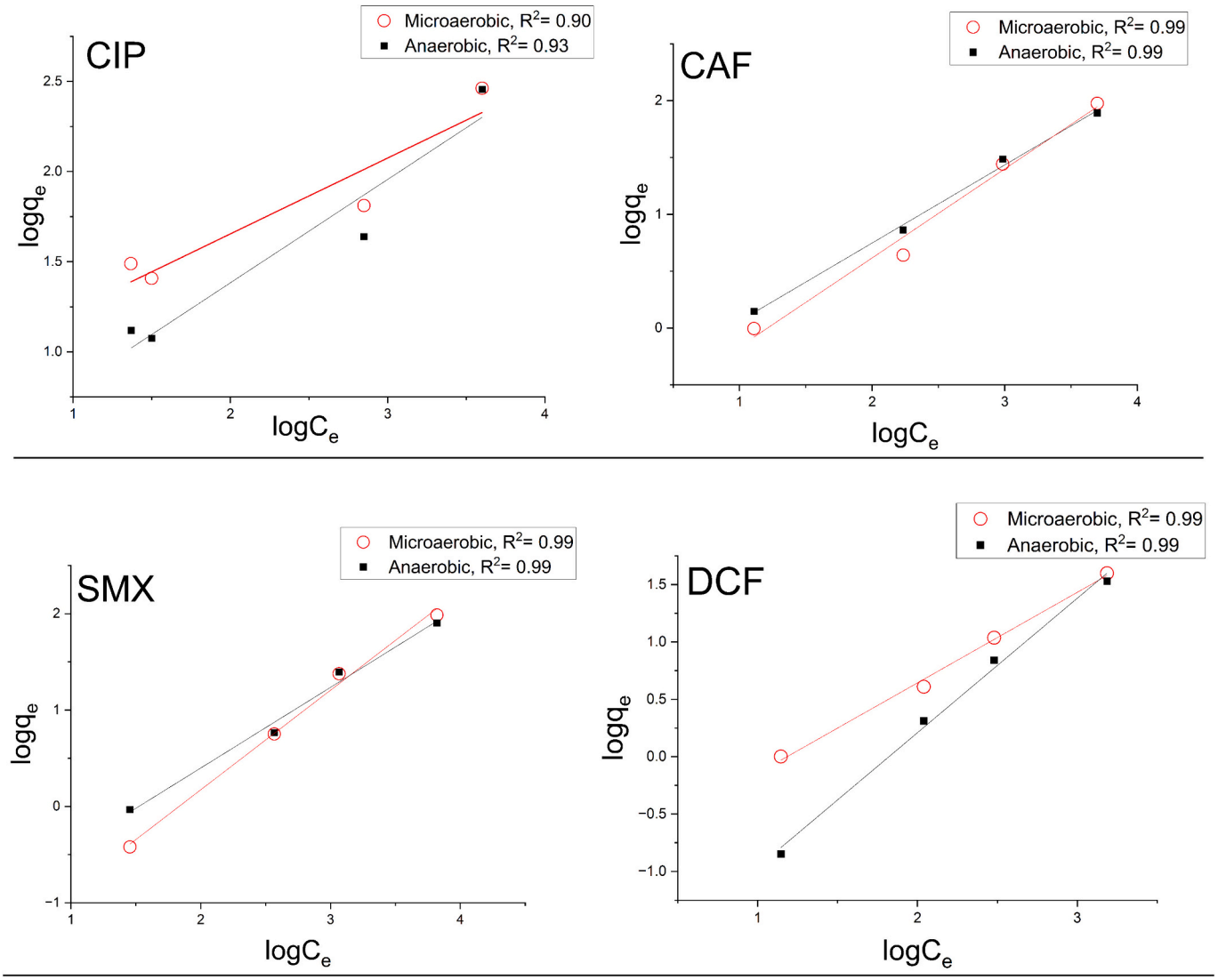
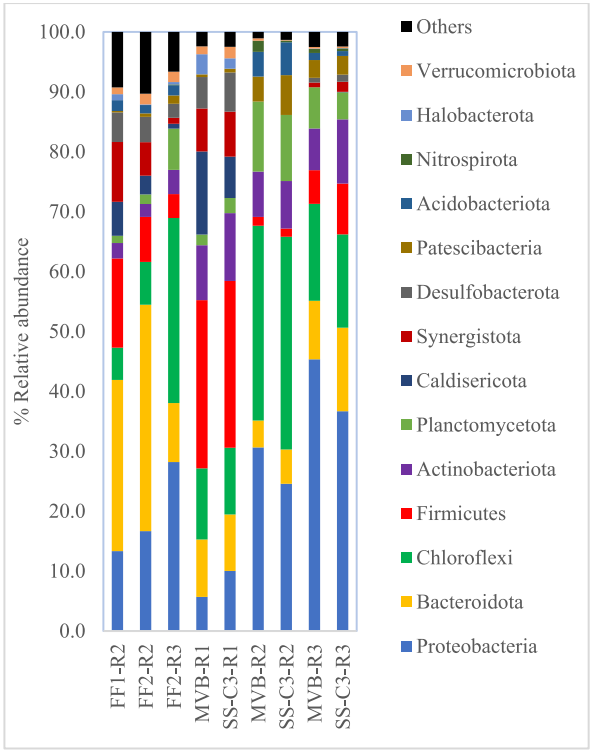


Fig. 7. Freundlich adsorption isotherms of targeted pharmaceuticals in batch adsorption tests between microaerobic and anaerobic conditions.

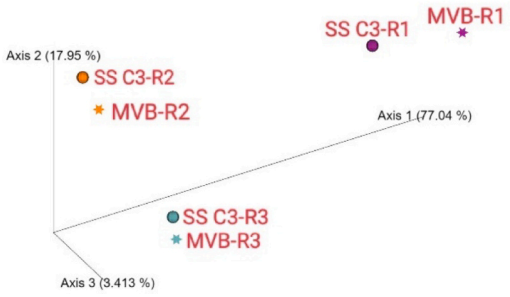
the Illumina MiSeq technique to analyze their microbial compositions. The effective readings of all samples were obtained at 16,682 reads. Fig. 8(a) shows the relative abundance of microbes at the phylum level. It was found that Firmicutes and bacteroidota are the main phylum in anaerobic biomass. Most of the Firmicutes engage in the synthesis of short-chain fatty acids (Walter et al., 2019). Chloroflexi and Proteobacteria are prominent categories of microorganisms that present in microaerobic biomass, contributing to the breakdown of organic substances and the elimination of nutrients (Dang et al., 2021). As shown in Fig. 8(b), PCoA was performed to assess the distinctions and resemblances among the microbial populations, which were present in the samples of both biofilm and suspended sludge. The degree of distinction within the microbial community was indicated by the percentages associated with each Axis (Axis 1, Axis 2, and Axis 3), with a longer relative distance indicating a lower level of similarity between the three samples (Jiang et al., 2021). The result shows that the microbial composition of the biofilms on the MVB and suspended sludges under microaerobic conditions (R2 and R3) were much closer to each other than was observed under anaerobic conditions (R1). This is likely due to the larger size and greater thickness of the biofilm and suspended flocs in the microaerobic environment. This condition closely resembles the characteristics of the biofilm. This is in contrast with the biofilm and suspended sludge in the anaerobic conditions, which had lower thickness due to having less density as also reported by Thoai et al. (2023).

A heat map of relative abundance in the genus level of the microbial communities as shown in Fig. 8(c) was created to gather specific information about diverse functional microorganisms, which involved the removal of pharmaceuticals under both anaerobic and microaerobic conditions. The heatmap shows that there was a significant difference among the microbial populations when the sole microaeration and the microaeration with sludge recirculation methods were applied in AnBB-MBR for both suspended and biofilm-associated microorganisms. Various microorganisms involved in nitrogen removal were also reported to be responsible for the biodegradation of the targeted pharmaceuticals in this work. Therefore, they might enhance the removal of targeted pharmaceuticals in the microaerobic R2 and R3 as compared to in the anaerobic R1. For instance, the co-metabolism and metabolism induced by Nitrosomonas and Nitrospira were reported to play a significant role in the CIP biodegradation in the nitrifying biofilm system (Dang et al., 2021). Nitrospira was reported as having strong tolerance for fluoroquinolone and ciprofloxacin, in the range of 0.9–9.0 mg/L (Meng et al., 2015).

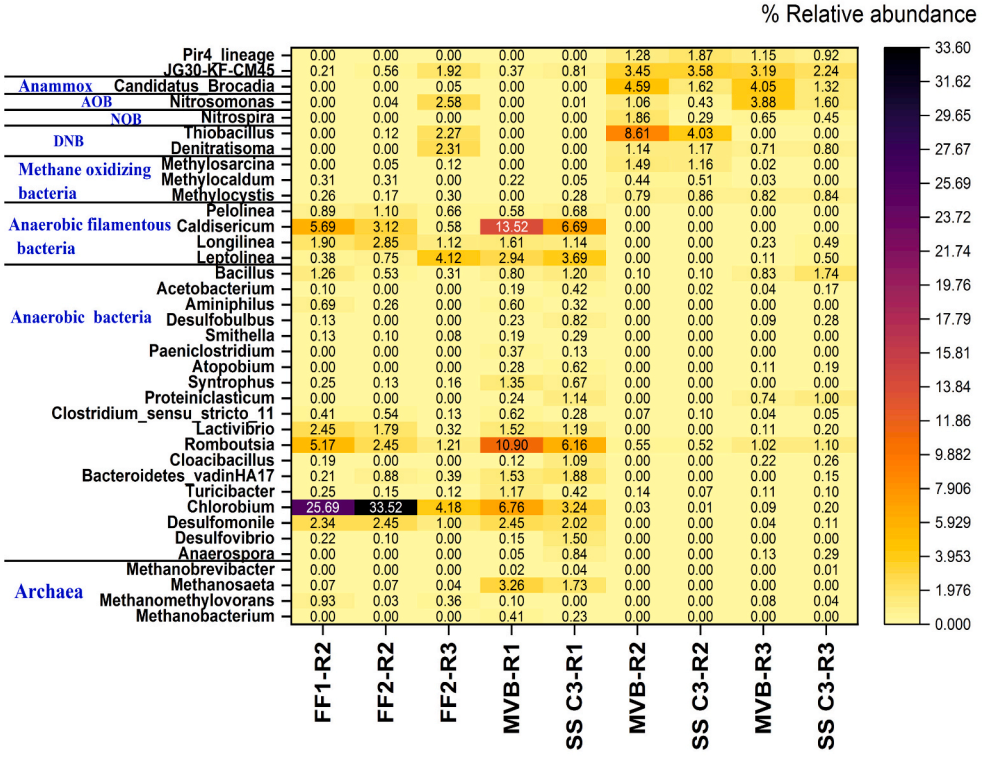
In addition, the ammonia monooxygenase enzyme induced by ammonia oxidizing bacteria (Nitrosomonas in this study) was also able to promote CAF and DCF co-metabolic biodegradation with nitrification



(a)



(b)



(c)

Fig. 8. Relative abundance at the phylum level (a); PCoA weighted Unifrac of the microbial community (b); Heat map of relative abundance at the genus level in the given samples of microorganisms (c).

activity as reported by Qiu et al. (2022) and Wang et al. (2022a). Kolakovic et al. (2022) demonstrated the correlation between the nitrifying activity and the increased biotransformation of DCF in the enhanced biological phosphorus removal process which might be associated with enhancing the removal of DCF under microaerobic conditions. The biotransformation of SMX was found to be highly influenced by the presence of nitrate-reducing conditions (denitrification) and it was also sensitive to nitrate concentration levels (de Wilt et al., 2018). In this case, Thiobacillus and Denitratisoma were observed to be as the major denitrifying bacteria in microaerobic biomasses (R2 and R3) but they were not observed in the anaerobic biomass (R1). Thiobacillus was also reported to play a role in the degradation of SMX (Hassan et al., 2020; Su et al., 2023). Furthermore, the exposure of SMX to partial nitrification in sequencing batch reactors led to an increase in the population of Denitratisoma, which could also enhance denitrifying activity (Li et al., 2023b). This reason might probably lead to enhancing the biodegradation of SMX in the microaerobic reactors (R2 and R3). Moreover, previous research has found that the increase of Denitratisoma, which is denitrifying bacteria, could promote complete denitrification under CIP exposure in a partial denitrification system (Li et al., 2023a).

Interestingly, relative abundances of Nitrosomonas (ammonia oxidizing bacteria, AOB) were found at 1.08 % and 3.86 % in MVB-R2 and MVB-R3, respectively, which were higher levels than in the suspended sludge samples at 0.43 % and 1.60 % in SS C3-R2 and SS C3-R3, respectively. Moreover, Candidatus brocadia (Anaerobic Ammonium Oxidation bacteria: Anammox) were also detected at higher levels in the biofilms in C3 of R2 and C3 of R3 at 4.59 % and 4.05 %, respectively than in the suspended sludges in C3 of R2 and C3 of R3 at 1.62 % and 1.32 %, respectively. These results are similar to those from previous research on a different system of anaerobic-aerobic bioreactor for treating high-strength ammonium piggery wastewater, in which it was found that Nitrosomonas and Candidatus brocadia thrived in the biofilm more than in the suspended sludges (Deng et al., 2019). This finding could confirm that the ammonium removal microorganisms in this study preferred to grow in biofilm-based treatment systems. Additionally, methane-oxidizing bacteria such as Methylosarcina, Methylocaldum, and Methylocystis were more abundant in the microaerobic biomasses (R2 and R3); however, this group was virtually absent in the anaerobic biomass. These microorganisms were able to break down SMX co-metabolically by stimulating the activity of methane monooxygenase enzymes ((Benner et al., 2015), while also removing dissolved methane from the R2 and R3 permeates more efficiently by 87.94 % and 100 % as compared to R1 (Fig. S2). This is in line with the findings of Wang et al. (2022), who found that Methylocaldum possesses the capability to biodegrade sulfonamide antibiotics. Also, Jechorek et al. (2003) reported that Methylocystis showed the capacity to break down aliphatic and chlorinated aromatic compounds.

Overall, the microbial community structure is considered to be a key contributing factor that could enhance the removal efficiency of the targeted compounds when microaeration at 0.93 L_{O2}/L_{feed} is employed in the AnBB-MBR system. For anaerobic microbes, Bacillus found at 0.80–1.20 % in the anaerobic biomass and at 0.10 % in the microaerobic biomass is associated with the biodegradation of antibiotics such as CIP and SMX (Fan et al., 2019; Feng et al., 2019). This might lead to biodegradation becoming the primary process in the degradation of CIP and SMX under anaerobic conditions. Moreover, Leptolinea and Longilinea, which are both members of the Anaerolineaceae family of strictly anaerobic filamentous bacteria, have the potential to produce the oxygenase enzymes, which could have participated in the cometabolism of the pharmaceuticals investigated in the present study.

The anaerobic filamentous microorganisms, which consist of the genus of Pelolinea (0.58–0.68 %), Caldisericum (6.69–13.52 %), Longilinea (1.14–1.61) and Leptolinea (3.94–3.69 %) were found in the anaerobic biomass of R1, whereas they were not found in the microaerobic biomass (R2) and found at only low levels in R3 biomass for Longilinea (0.23–0.49 %) and Leptolinea (0.11–0.50 %). This is in accordance with SEM images, which show that anaerobic filamentous bacteria were enriched on the MVB biofilms of R1 and R3. These microorganisms have a negative impact on membrane filtration by inducing significant levels of L-EPS (Fig. 6(b) and (c)), which exhibited high hydrophobicity, formed irregular flocs, and had a low zeta potential. These microorganisms also had adverse effects on membrane filtration within the anaerobic MBR system (Meng et al., 2006), which is consistent with results from the AnBB-MBR (R1) in this study. Even with the existence of these microorganisms in R3, there was still sludge recirculation that could alleviate membrane fouling by inducing shear force on to the foulants by the moving MVB. In addition, Desulfobrio, found at 0.15–1.50 % in the anaerobic biomass and not found in the microaerobic biomass, likely influenced the accumulation of organic fouling materials (Yao et al., 2021), thereby affecting the dynamics of the biocakes of R1. For other microbial groups, JG30-KF-CM45 are significant contributors to the breakdown of complex organic molecules in wastewater treatment plants (Huang et al., 2022). As these groups are more prevalent in the microaerobic biomass compared to the anaerobic biomass of R1, they might be crucial contributors to the removal of complex organic substances and foulants from the inlet building wastewater. Also, Pir4_lineage is associated with the microaerobic microbes that have been detected in anoxic or micro-oxic habitats (Dedysh et al., 2020).

4. Conclusion

The proposed AnBB-MBR with microaeration (R2) and AnBB-MBR with microaeration and sludge recirculation (R3) were employed in AnBB-MBR (R1) to enhance the removal of CIP, CAF, SMX and DCF from building wastewater. R2 and R3 showed notable enhancements in the removal of COD, VFAs, and dissolved methane, as well as in the reduction of the membrane fouling rate when compared to R1. C1 in the AnBB-MBRs were most responsible for the removal of all the targeted compounds. The highest removal of CIP (78.0 %) and DCF (40.8 %) occurred in R2, while the highest removal of SMX was in R3 at 91.3 %; whereas CAF was efficiently removed (>90 %) in all of the AnBB-MBR systems. Microaeration enhanced the adsorption of CIP onto the biomass and assisted the biodegradability of CAF, SMX and DCF in the AnBB-MBR, which were insisted by batch biotransformation kinetics and adsorption tests. Microaeration was found to be able to enrich the microbial groups that were responsible for the degradation of the targeted pharmaceuticals. Overall, the proposed AnBB-MBR with microaeration is a promising solution for the treatment of building wastewater contaminated with pharmaceuticals.

CRediT authorship contribution statement

Tanissorn Buakaew: Writing – original draft, Validation, Data curation, Conceptualization. **Chavalit Ratanatamskul:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2024.172420.

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